Qualcomm Technologies, Inc.

00068.1 Release Note for SW6100.LAW.1.0

Release Notes

SW6100.LAW.1.0

RNO-260408013424 Rev. 1

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About Qualcomm

Qualcomm relentlessly innovates to deliver intelligent computing everywhere, helping the world tackle some of its most important challenges. Building on our 40 years of technology leadership in creating era-defining breakthroughs, we deliver a broad portfolio of solutions built with our leading-edge AI, high-performance, low-power computing, and unrivaled connectivity. Our Snapdragon® platforms power extraordinary consumer experiences, and our Qualcomm Dragonwing™ products empower businesses and industries to scale to new heights. Together with our ecosystem partners, we enable next-generation digital transformation to enrich lives, improve businesses, and advance societies. At Qualcomm, we are engineering human progress.

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1. Download

A. ChipCode Location:

https://chipcode.qti.qualcomm.com/lg-electronics-a-k-a-lge/sw6100-law-1-0_amss_oem_osh

https://chipcode.qti.qualcomm.com/lg-electronics-a-k-a-lge/sw6100-law-1-0_amss_standard_oem

https://chipcode.qti.qualcomm.com/lg-electronics-a-k-a-lge/sw6100-law-1-0_amss_standard_oem_nhlos

https://chipcode.qti.qualcomm.com/lg-electronics-a-k-a-lge/sw6100-law-1-0_ap_oem_osh

https://chipcode.qti.qualcomm.com/lg-electronics-a-k-a-lge/sw6100-law-1-0_ap_standard_oem

https://chipcode.qti.qualcomm.com/lg-electronics-a-k-a-lge/sw6100-law-1-0_hlos_device_oss

https://chipcode.qti.qualcomm.com/lg-electronics-a-k-a-lge/sw6100-law-1-0_hlos_dev_wear-hal

https://chipcode.qti.qualcomm.com/lg-electronics-a-k-a-lge/sw6100-law-1-0_test_device

B. ChipCode Commands:

git clone --depth 1 <repo location>
Please note that the <repo location> and clone path can be obtained from the "Files" tab in the "Clone this repository" section of the above chipcode location.

2. Release History

ChipCode Release History:
https://createpoint.qti.qualcomm.com/planner/link/search/newReleaseHistory/preset/sw6100-law-1-0_amss_oem_osh/master/r00068.1

https://createpoint.qti.qualcomm.com/planner/link/search/newReleaseHistory/preset/sw6100-law-1-0_amss_standard_oem/master/r00068.1

https://createpoint.qti.qualcomm.com/planner/link/search/newReleaseHistory/preset/sw6100-law-1-0_amss_standard_oem_nhlos/master/r00068.1

https://createpoint.qti.qualcomm.com/planner/link/search/newReleaseHistory/preset/sw6100-law-1-0_ap_oem_osh/master/r00068.1

https://createpoint.qti.qualcomm.com/planner/link/search/newReleaseHistory/preset/sw6100-law-1-0_ap_standard_oem/master/r00068.1

https://createpoint.qti.qualcomm.com/planner/link/search/newReleaseHistory/preset/sw6100-law-1-0_hlos_device_oss/master/r00068.1

https://createpoint.qti.qualcomm.com/planner/link/search/newReleaseHistory/preset/sw6100-law-1-0_hlos_dev_wear-hal/master/r00068.1

https://createpoint.qti.qualcomm.com/planner/link/search/newReleaseHistory/preset/sw6100-law-1-0_test_device/master/r00068.1

3. Getting Started

Software maturity of SP

Software product	Maturity level
SW6100.LAW.1.0	POST-CS1

Download QTI proprietary software from Qualcomm ChipCode portal

Qualcomm Technologies, Inc. (QTI) software can be downloaded from the Qualcomm ChipCode website. Designated points of contact in your organization can download the licensed software. The software is organized into distribution packages (distros) composed of subsystem image files. Each distro has a corresponding Git project. The Git tree includes revisions for previous builds that allow you to diff the changes between releases.

If you are accessing Qualcomm ChipCode for the first time, QTI recommends reviewing the up-to-date documentation and tutorial videos at the ChipCode Help Center: https://chipcode.qti.qualcomm.com/helpki.

Follow steps below to download the QTI proprietary software.

1. Go to the ChipCodeTM portal by clicking the link: https://chipcode.qti.qualcomm.com

2. Select <SW6100.LAW.1.0> product from the Products list. Distributions (distros) available will be listed.

3. Choose a distro to see the available release tags. The release tag closely matches the build ID in the release notes. For this release:

    - Build ID: POST-CS1 0.0.068.1

    - Chipcode tag: r00068.1

4. Download the distro from the Qualcomm ChipCode website by using one of the following methods:

    - **Note**: The about.html and contents.xml files contain the build ID used to download the open-source software from CodeLinaro sites

- Method 1:
  - On the distro page, select a release tag and use the Download Release button to download in GZip (Linux) or Zip (Windows) format.
  - Create a top-level local directory on the build PC and decompress the downloaded software package.
- Method 2: On the distro page, choose the <release-tag> execute the command in the Clone this release box to clone the release with git, and then check out the specific release tag, following is an example of the git commands:
  - Shallow clone
    - ```
      $ git clone -b r00068.1 --depth 1 https://qpm-git.qualcomm.com/home2/git/qualcomm/sw6100-law-1-0_amss_standard_oem.git
      ```
  - Full clone
    - ```
      $ git clone https://qpm-git.qualcomm.com/home2/git/qualcomm/sw6100-law-1-0_amss_standard_oem.git
      ```

  Note: A shallow clone copies only the specific release. A full clone is performed to copy all available releases, but may result in large transfers and possible connection drops.

Request customer access to CodeLinaro (CLO)

There are two types of CLO: Private CLO and Public CLO. Currently, Qualcomm is using private CLO to maintain the opensource code for this SP.

Private CLO access requires a CLO account.

- One time CLO set-up is required, see “Set up CodeLinaro” section in the Quick Start Guide (80-35346-1).

- If one does not have a personal access token, generate one by following “3.4.1 Generate Personal Access Tokens” section in 80-35346-1.

Private CLO path for this SP:

- clo/private/law-vendor-16-0

- clo/private/la-qxsi-16-0/

- clo/private/kernel-platform-5-0

- clo/private/kernel-devicetree

Android Build Informations:

Date	Tag/ BuildId	Android Version
Apr 6, 2025	KERNEL: AU_LINUX_KERNEL.PLATFORM.5.0.R5.00.00.00.130.074 GRAPHICS: AU_TECHPACK_GRAPHICS.LA.16.1.R1.00.00.00.000.043 SENSORS: AU_TECHPACK_SENSORS.AP.1.0.2.R3.00.00.00.000.027 BTFM: AU_TECHPACK_BTFM.WEAR.1.0.R1.00.00.00.000.046 WLAN: AU_TECHPACK_WLAN.LW.1.0.R1.00.00.00.000.059 AUDIO: AU_TECHPACK_AUDIO_IOT.LA.11.0.R1.00.00.00.000.078 DISPLAY: AU_TECHPACK_DISPLAY.LA.6.10.R1.00.00.00.000.091 VIDEO: AU_TECHPACK_VIDEO.LA.2.4.R2.00.00.00.000.045 VENDOR: AU_LINUX_ANDROID_LAW.VENDOR.16.0.R1.11.00.00.1345.188 QXSI: AU_LINUX_ANDROID_LA.QXSI.16.0.R1.11.00.00.1341.110	**Android W**

Keystone Tag: BQ2A.260219.001
Android Upstream Tag: BP2A.250705.008

4. Build Instructions

Metabuild components:

New Images: LPAI.WM.1.2

Obsolete Images:
LPAI.WM.1.1/DCP.FW.1.1/SMC.THEMISTO.1.0.1/TMEL.THEMISTO.1.1/BTFW.THEMISTO.1.0.0/WLAN.LW.1.0.r1/BTFW.THEMISTO.2.0.0/BOOT.MXF.2.5/TZ.XF.5.0

Note?DCP.FW.1.1 SI is obsolete; the dcp.bin has been moved to the LAW.VENDOR.16.0.r1 vendor image.

Software Image	Build ID
Meta	**SW6100.LAW.1.0.r1-00068-STD.PROD-1**
ACPOLICY.XF.1.0	ACPOLICY.XF.1.0-00558-ASPEN-1
AOP.HO.6.0	AOP.HO.6.0-00347-ASPEN_E-1
BOOT.MXF.2.5.6	BOOT.MXF.2.5.6-00050-ASPEN-1
BTFW.THEMISTO.2.0.1	BTFW.THEMISTO.2.0.1-00098-PATCHZ-1
CDSP.HT.3.4	CDSP.HT.3.4-00273-ASPEN-1
CPUCP.FW.1.0	CPUCP.FW.1.0-00017-ASPEN.EXT-1
LPAI.WM.1.2	LPAI.WM.1.2-00054-LPAI_WM_PACK-1
LPAIDSP.HT.1.4	LPAIDSP.HT.1.4-00775-ASPEN-1
MPSS.CY.2.0	MPSS.CY.2.0-00671-ASPEN_GEN_PACK-1
SMC.THEMISTO.2.0.1	SMC.THEMISTO.2.0.1-00102-STD.SMC.PROD-1
TMEL.THEMISTO.2.0	TMEL.THEMISTO.2.0-00042-PT_GEN_PACK-2
TZ.APPS.1.0	TZ.APPS.1.0-02126-ASPEN-1
TZ.XF.5.35	TZ.XF.5.35-00005-ASPEN-1
WLAN.MN.2.0	WLAN.MN.2.0-01664-QCAMNSWPLZ-1
KERNEL.PLATFORM.5.0.r5	KERNEL.PLATFORM.5.0.r5-07400-kernel.0-1
GRAPHICS.LA.16.1.r1	GRAPHICS.LA.16.1.r1-04300-SW6100.0-1
SENSORS.AP.1.0.2.r3	SENSORS.AP.1.0.2.r3-02700-SW6100.0-1
BTFM.WEAR.1.0.r1	BTFM.WEAR.1.0.r1-04600-SW6100.0-1
WLAN.LW.1.0.r1	WLAN.LW.1.0.r1-05900-SW6100.0-2
AUDIO_IOT.LA.11.0.r1	AUDIO_IOT.LA.11.0.r1-07800-SW6100.0-1
DISPLAY.LA.6.10.r1	DISPLAY.LA.6.10.r1-09100-SW6100.0-1
VIDEO.LA.2.4.r2	VIDEO.LA.2.4.r2-04500-SW6100.0-1
LA.QXSI.16.0.r1	LA.QXSI.16.0.r1-11000-qssi.0-1
LAW.VENDOR.16.0.r1	LAW.VENDOR.16.0.r1-18800-SW6100.0-1

For the build components and IDs, refer to the ***about.html*** file in the root of the repository on Qualcomm ChipCode

4.1 Download and Build HLOS software images

Prerequisites:

- The HLOS proprietary code is downloaded from the ChipCode portal(Followed the Chapter 3).
- Set build environment(Supported build hosts Ubuntu 22.04):

```
$ sudo apt-get install libstdc++-12-dev
```

```
$ sudo apt-get install libxml-simple-perl
```

4.1.1 Sample Commands to download & generate different workspace combinations

- **LA.QXSI.16 STANDALONE TREE (CLO + Proprietary content from Qualcomm ChipCode):**

```
$ mkdir -p <qxsi_workspace_path>
$ cd <qxsi_workspace_path>
$ cp <absoulte path where chipcode
extracted>/LA.QXSI.16.0/LINUX/android/sync_snap_v2.sh ./
$ ./sync_snap_v2.sh \
  --workspace_path=<qxsi_workspace_path> \
  --image_type=la \
  --tree_type=la_qssi  \
  --prop_opt=chipcode \
  --common_oss_url=https://git.codelinaro.org \
  --qssi_oss_manifest_git=clo/private/la-
qxsi-16-0/la/system/manifest \
  --
qssi_au_tag=AU_LINUX_ANDROID_LA.QXSI.16.0.R1.11.00.00.1341.110
\
  --qssi_chipcode_path=<absoulte path where chipcode
extracted>/LA.QXSI.16.0/LINUX/android \
  --repo_url=https://git.codelinaro.org/clo/tools/repo.git \
  --repo_branch=aosp-new/stable \
  --shallow_clone=true
```

After downloading, please verify whether the directory
`<qxsi_workspace_path>/vendor/qcom/proprietary` exists.
If it does not exist, manually copy it using the following command:
cp -a
<absolute_path_where_chipcode_extracted>/LA.QXSI.16.0/LINUX/android/vendor/qcom/proprietary <qxsi_workspace_path>/vendor/qcom/

- **For Compilation of QXSI 16:**

```
$ source build/envsetup.sh
$ lunch qssi_wear-userdebug
$ bash build.sh -j32 dist --qssi_only
```

- KERNEL.PLATFORM STANDALONE TREE (CLO + Proprietary content from Qualcomm ChipCode):

```
$ mkdir -p <kernel_workspace_path>
$ cp <absoulte path where chipcode
extracted>/KERNEL.PLATFORM.5.0/kernel_platform/sync_snap_v2.sh
./
$ cd <kernel_workspace_path>
$ ./sync_snap_v2.sh \
  --image_type=la \
  --tree_type=kernel_platform \
  --prop_opt=chipcode \
  --workspace_path=<kernel_workspace_path> \
  --common_oss_url=https://git.codelinaro.org \
  --kernel_oss_manifest_git=clo/private/kernel-
platform-5-0/kernelplatform/manifest \
  --
kernel_au_tag=AU_LINUX_KERNEL.PLATFORM.5.0.R5.00.00.00.130.074
\
  --kernel_chipcode_path=<absoulte path where chipcode
extracted>/KERNEL.PLATFORM.5.0/kernel_platform \
  --repo_url=https://git.codelinaro.org/clo/tools/repo.git \
  --repo_branch=aosp-new/stable \
  --shallow_clone=true
```

- For Compilation of KERNEL:

```
$ cd <kernel_workspace_path>
$ cd kernel_platform
$ ./build_with_bazel.py -t vienna consolidate
```

- VENDOR TREE (CLO + Proprietary content from Qualcomm ChipCode):

```
$ mkdir -p <vendor_techpack_workspace_path>
$ cd <vendor_techpack_workspace_path>
$ cp <absoulte path where chipcode
extracted>/LAW.VENDOR.16.0/LINUX/android/sync_snap_v2.sh ./
$ ./sync_snap_v2.sh \
  --workspace_path=<vendor_techpack_workspace_path> \
  --image_type=la \
  --prop_opt=chipcode \
```

```
  --tree_type=la_vendor_techpack \
  --common_oss_url=https://git.codelinaro.org \
  --vendor_oss_manifest_git=clo/private/law-
vendor-16-0/la/vendor/manifest \
  --
vendor_au_tag=AU_LINUX_ANDROID_LAW.VENDOR.16.0.R1.11.00.00.1345
.188 \
  --vendor_chipcode_path=<absoulte path where chipcode
extracted>/LAW.VENDOR.16.0/LINUX/android \
  --display_oss_manifest_git=clo/private/law-
vendor-16-0/techpack/display/manifest \
  --
display_au_tag=AU_TECHPACK_DISPLAY.LA.6.10.R1.00.00.00.000.091
\
  --display_chipcode_path=<absoulte path where chipcode
extracted>/DISPLAY.LA.6.10/LINUX/android \
  --video_oss_manifest_git=clo/private/law-
vendor-16-0/techpack/video/manifest \
  --video_au_tag=AU_TECHPACK_VIDEO.LA.2.4.R2.00.00.00.000.045 \
  --video_chipcode_path=<absoulte path where chipcode
extracted>/VIDEO.LA.2.4/LINUX/android \
  --audio_oss_manifest_git=clo/private/law-
vendor-16-0/techpack/audio/manifest \
  --
audio_au_tag=AU_TECHPACK_AUDIO_IOT.LA.11.0.R1.00.00.00.000.078
\
  --audio_chipcode_path=<absoulte path where chipcode
extracted>/AUDIO_IOT.LA.11.0/LINUX/android \
  --sensor_oss_manifest_git=clo/private/law-
vendor-16-0/techpack/sensors/manifest \
  --
sensor_au_tag=AU_TECHPACK_SENSORS.AP.1.0.2.R3.00.00.00.000.027
\
  --sensor_chipcode_path=<absoulte path where chipcode
extracted>/SENSORS.LA.6.0/LINUX/android \
  --graphics_oss_manifest_git=clo/private/law-
vendor-16-0/techpack/graphics/manifest \
  --
graphics_au_tag=AU_TECHPACK_GRAPHICS.LA.16.1.R1.00.00.00.000.04
3 \
  --graphics_chipcode_path=<absoulte path where chipcode
extracted>/GRAPHICS.LA.16.1/LINUX/android \
```

```
  --wlan_oss_manifest_git=clo/private/law-vendor-16-0/techpack/wlan/manifest \
  --wlan_au_tag=AU_TECHPACK_WLAN.LW.1.0.R1.00.00.00.000.059 \
  --wlan_chipcode_path=<absoulte path where chipcode extracted>/WLAN.LW.1.0/LINUX/android \
  --btfm_oss_manifest_git=clo/private/law-vendor-16-0/techpack/btfm/manifest \
  --btfm_au_tag=AU_TECHPACK_BTFM.WEAR.1.0.R1.00.00.00.000.046 \
  --btfm_chipcode_path=<absoulte path where chipcode extracted>/BTFM.WEAR.1.0/LINUX/android \
  --repo_url=https://git.codelinaro.org/clo/tools/repo.git \
  --repo_branch=aosp-new/stable \
  --shallow_clone=true
```

- **Copy the content from QSSI & Kernel workspace:**

```
$ cp <vendor_techpack_workspace_path>/vendor/qcom/proprietary/prebuilt_HY11/Android.mk <Backup_path>/Android.mk
$ cp -a <qxsi_workspace_path>/* <vendor_techpack_workspace_path>/
$ cp -a <kernel_workspace_path>/* <vendor_techpack_workspace_path>/
$ cp <Backup_path>/Android.mk <vendor_techpack_workspace_path>/vendor/qcom/proprietary/prebuilt_HY11/Android.mk
```

- **For Compilation of Vendor:**

```
$ source build/envsetup.sh
$ lunch vienna64-userdebug
$ chmod 777 kernel_platform/build/android/prepare_vendor.sh
$ chmod 777 kernel_platform/build/build_module.sh
$ chmod 777 kernel_platform/build/android/merge_dtbs.sh
$ RECOMPILE_KERNEL=1 ./kernel_platform/build/android/prepare_vendor.sh vienna consolidate </dev/null
$ chmod 777 ./build.sh
$ ./build.sh -j32 dist --target_only
```

- Generate super.img

```
$ python vendor/qcom/opensource/core-
utils/build/build_image_standalone.py \
  --qssi_build_path <qxsi_workspace_path> \
  --target_build_path <vendor_techpack_workspace_path> \
  --merged_build_path <vendor_techpack_workspace_path> \
  --target_lunch vienna64 \
  --output_ota \
  --image super
```

4.2 Build non-HLOS software images

4.2.1 Toolchain

The tables list the toolchain and version dependencies for individual non-HLOS components, which should be installed on the build machine.

Table 5-1 Non-HLOS toolchain to build source for SW6100.LAW.1.0

Component	Source or Binary	Toolchain for building source	Python Version	Perl Version	Prerequisites	Build hosts
AOP.HO.6.0	Source	Snapdragon® LLVM RISCV Toolchain OEM 17.0.0	Python 3.8.2	–	–	Ubuntu 22.04
BOOT.MXF.2.5	Source	Snapdragon® LLVM Toolchain for Arm® technology 18.0.1 Snapdragon® LLVM Toolchain for Arm® technology 19.0.0 DTC 1.6.0	Python 3.8.2	–	–	Ubuntu 22.04
CDSP.HT.3.4	Source	Hexagon tools 19.0.05 DTC 1.6.0	Python 3.12.7	–		Ubuntu 22.04
LPAIDSP.HT.1.4	Source	Hexagon tools 19.0.05 Snapdragon® LLVM Toolchain for Arm® technology 18.0.1 DTC 1.6.0	Python 3.12.7	–	Nanopb version 0.4.8	Ubuntu 22.04
MPSS.CY.2.0	Source	Hexagon 8.7.04	Python 3.8.2	Perl 5.18.2+	Nanopb version 0.4.8	Ubuntu 22.04
LPAI.WM.1.1	Source	Zephyr Version: 3.7 LTS Zephyr SDK Version: 0.16.x ARM GCC 16.1.0 SECTOOLS	Python 3.12.10		Zephyr Getting Started Guide (https://docs.zephyrproject.org/latest/develop/getting_started/index.html)	Ubuntu 22.04
TZ.XF.5.0	Source	LLVM 18.0.1	Python 3.8.2 or Python 3.10.2	–	–	Ubuntu 22.04
TZ.APPS.1.0	Binary	–	–	–	–	–
CPUCP.FW.1.0	Binary	–	–	–	–	–
DCP.FW.1.1	Binary	–	–	–	–	–
BTFW.THEMISTO.1.0.0	Binary	–	–	–	–	–
SMC.THEMISTO.1.0.1	Binary	–	–	–	–	–
WLAN.MN.1.0	Binary	–	–	–	–	–
TMEL.THEMISTO.1.1	Binary	–	–	–	–	–

4.2.2 Set build environment

Install required packages

You need to install the required packages on the host machine before starting to compile. The example commands for those installations as below.

1. Run these commands to install necessary software packages:

```
$ sudo apt install python3-pip uuid-dev libffi-dev libxml-parser-perl flex
```

2. Install Python 3.8.2 and Python 3.12.7.

3. Install additional packages for Python

```
$ sudo pip3 install pyfdt
$ sudo pip3 install future
$ sudo pip3 install dtschema==2021.10
```

Install Device Tree Compiler (DTC)

It is recommended that the OEM installs DTC from the source to ensure the installed version has the YAML output support, which is required for dt-schema.

Follow thest steps to install DTC from Linux source:

1. Download DTC v1.6.0 with:

```
$ git clone -b v1.6.0 https://git.kernel.org/pub/scm/utils/dtc/dtc.git
```

2. Make sure **libyaml-dev** is installed.

3. Run `make` at the top-level of DTC code tree. DTC is generated at the top-level of tree.

4. Run `sudo make CFLAGS=-Wno-error install` to install DTC.

Other

The reference commands enable you to set environment variables for toolchains of all non-HLOS components. Not using these environment variables does not impact the build. The environment variables specific to a non-HLOS component can be set. For information on tool dependency and version, see Table 5-1 Non-HLOS toolchain to build source for SW6100.LAW.1.0. Accordingly, set the required environment variable.

Ensure that PYTHON_PATH and SECTOOLS variables are set for all the non-HLOS components.

- **Linux build environment**

```
$ export PYTHON_PATH=<path_to_python>/bin

$ export HEXAGON_ROOT=<path_to_hexagon_tool_install_folder>

$ export
SECTOOLS=<target_root>/SW6100.LAW.1.0/common/sectoolsv2/ext/Linux/sectools

$ export DTC_PATH=<path_to_DTC_tool_install_folder>
```

4.2.3 Build MPSS

For APQ distro **sw6100-law-1-0_ap_standard_oem**, the MPSS compilation is not required .

Set environment variables for MPSS build. For more information, see Set build environment.

Setup nanopb

nanopb is automatically downloaded as part of the MPSS build. If there is an error downloading nanopb at build time, then nanopb can be downloaded manually.

Linux build environment

1. `$ cd <target_root>/MPSS.CY.2.0/modem_proc/qsh_api`
2. `$ wget https://jpa.kapsi.fi/nanopb/download/nanopb-0.4.8-linux-x86.tar.gz`
3. `$ cd <target_root>/MPSS.CY.2.0/modem_proc`
4. `$ python qsh_api/build/config_nanopb_dependency.py -f nanopb-0.4.8-linux-x86`
5. `$ cd <target_root>/MPSS.CY.2.0`

6. Run build command from the table

Build environment	Build commands
Linux	◦ Build images: `$ ./modem_proc/build/ms/build.sh aspen.gen.prod`

4.2.3 Build boot loader

Set environment variables for boot build. For more information, see Set build environment. Toolchain path can be updated in the file "BOOT.MXF.2.5/boot_images/boot/toolchainconfig.json".

1. `$ export SECTOOLS_DIR=<target_root>/SW6100.LAW.1.0/common/sectoolsv2/ext/Linux`

2. Go to the following directory: $ `$ cd <target_root>/BOOT.MXF.2.5.6`

   - Where **<target_root>** is the top-level directory created in Download QTI proprietary software

3. Run build command from the table

Build environment	Build command
Linux	◦ Build images `$ python -u boot_images/boot_tools/buildex.py -t Aspen,QcomToolsPkg,QcomTestPkg -r RELEASE -v LW`

4.2.4 Build TZ and hypervisor

Set environment variables for TZ build. For more information, see Set build environment

1. unset SECTOOLS_DIR

2. Go to the following directory

```
$ cd <target_root>/TZ.XF.5.35/trustzone_images/build/ms
```

3. Run build command from the table to build all images. The TZ and hypervisor images are released as binaries. Sampleapp and devcfg are generated with the following commands.

Build environment	Build command

Linux	◦
	Build images:
	`$ python build_all.py -b TZ.XF.5.0 CHIPSET=aspen`
	◦
	Clean the build:
	`$ python build_all.py -b TZ.XF.5.0 CHIPSET=aspen --clean`

4.2.5 Build AOP

Set environment variables for RPM build. For more information, see Set build environment.

1. Go to the following directory

```
$ cd <target_root>/AOP.HO.6.0/aop_proc/build
```

2. Run build command from the table

Build environment	Build command

Linux	◦
	Build images:
	`$ ./build_aspen_devcfg.sh`
	◦
	Clean the build
	`$ ./build_aspen_devcfg.sh –c`

4.2.6 Build ADSP

Set environment variables for ADSP build. For more information, see Set build environment.

Linux build environment

1. $ export SECTOOLS_DIR=**<target_root>**/SW6100.LAW.1.0/common/sectoolsv2/ext/Linux
2. $ cd **<target_root>**/LPAIDSP.HT.1.4/adsp_proc/qsh_api
3. $ wget https://jpa.kapsi.fi/nanopb/download/nanopb-0.4.8-linux-x86.tar.gz
4. $ cd ..
5. $ python qsh_api/build/config_nanopb_dependency.py -f nanopb-0.4.8-linux-x86
6. $ cd bt/tools
7. $ python bt_nanopb_dependency.py -f ../../qsh_api/nanopb-0.4.8-linux-x86.tar.gz
8. $ cd ../../build/ms/
9. Run build command from the table

Build environment	Build command

Linux	◦ Build images:
	$ python build_variant.py aspen.adsp.prod
	◦ Clean the build
	$ python ./build_variant.py aspen.adsp.prod --clean

4.2.7 CDSP

Set environment variables for ADSP build. For more information, see Set build environment.

Linux build environment

1. $ cd **<target_root>**/CDSP.HT.3.4/cdsp_proc/build/ms

2. Run build command from the table

Build environment	Build command

Linux	◦ Build images:
	$ python build_variant.py aspen.cdsp.prod
	◦ Clean the build
	$ python ./build_variant.py aspen.cdsp.prod --clean

4.2.8 LPAI.WM

Set environment variables for LPAI.WM build. For more information, see Set build environment.

Linux build environment

1. Install python 3.12.10

2. Refer to the **Zephyr** Getting Started Guide — Zephyr Project Documentation to configure the needed tools, SDK, and toolchains. Ensure to install the Zephyr SDK version **0.16.9.**

3. Browse to **<target_root>**/LPAI.WM.1.2/wm_proc/modules/nanopb. Create the nanopb directory if not existing already. Clone GitHub - zephyrproject-rtos/nanopb: Protocol Buffers with small code size into the folder

4. $ git checkout f2afdd7a8e8a9556d1c454acc91ec632fe69463f. This commands gets the same version that was used during internal testing.

5. sudo apt install protobuf-compiler

6. $ export PYTHONBIN=**<path_to_python>**/bin

7. $ export ZEPHYR_SDK_INSTALL_DIR=**<path_to_ZEPHYR_SDK>**

8. pip install protobuf grpcio-tools

9. $ cd **<target_root>**/LPAI.WM.1.2/wm_proc/build/ms/

10. Run build command from the table

Build environment	**Build command**

Linux	◦
	Build images:
	$ python ./build_variant.py aspen.swm.prod
	$ python ./build_variant.py aspen.awm.law_prod
	◦
	To clean
	$ python ./build_variant.py aspen.swm.prod --clean
	$ python ./build_variant.py aspen.awm.law_prod --clean

4.2.9 Others

TZ.APPS/CPUCP.FW/BTFW.THEMISTO/SMC.THEMISTO/WLAN.MN/TMEL.THEMISTO images is released as a binary and build compilation is not needed.

4.3 Generate META build

1. Before comping the META, please copy all HLOS images:

   $ cd /LAW.VENDOR.16.0/LINUX/android/

   $ mkdir -p out/target/product device/qcom/vienna-kernel

   $ cp -rvf **<HLOS_BUILD>**/out/target/product/vienna64 **<target_root>**/LAW.VENDOR.16.0/LINUX/android/out/target/product/vienna64

```
$ cp -rvf <HLOS_BUILD>/device/qcom/vienna-kernel/vmlinux
<target_root>/LAW.VENDOR.16.0/LINUX/android/device/qcom/vienna-kernel/vmlinux
```

```
$ cp -rvf <HLOS_BUILD>/out/host/linux-x86 <target_root>/LAW.VENDOR.16.0/LINUX/android/out/host/linux-x86
```

2. Go to the directory:

```
$ cd <target_root>/SW6100.LAW.1.0/common/build
```

3. Enter the command: (Python version to be used here is 3.8.2)

```
$ python build.py
```

4.4 CDT Infos

Change the platform ID based on hardware reference design.

For example, for SW6100.LAW.1.0, the supported platform IDs are: CDT Version:5,Platform ID:0x24,Major ID:0x01,Minor ID:0x00,Subtype:0x00

Device	Desc.	version	Platform Type (8 bit)	Minor Hardware Version (8 bit)	Major Hardware Version (8 bit)	Sub-type (8 bit)
WDP	Default and Lando+Pandeiro	0x05	0x24	0x01	0x00	0x00
WDP	Default and Landeiro	0x05	0x24	0x02	0x00	0x00
WDP	No Camera and Lando+Pandeiro	0x05	0x24	0x01	0x00	0x01
WRD	Default and Lando+Pandeiro	0x05	0x2d	0x01	0x00	0x00
WRD	Default and Landeiro	0x05	0x2d	0x02	0x00	0x00
WRD	No Camera and Lando+Pandeiro	0x05	0x2d	0x01	0x00	0x01

Please refer to SW6100 Software User Guide (80-74841-50) for CDT file generation & program.

5. Memory configuration

Memory Map Info (MSM):

This software can be used with the SW6100 ASICs and revisions, with the indicated release quality. ASIC revisions available at the time of this release are assumed to be supported, unless otherwise indicated.

The following figure shows the run-time memory map for the SW6100 ASICs.

NOTE: This is a preliminary memory map and may change in the future releases. It is for reference-only in this release and should not be used for optimization.

PIL Memory Configuration and Usage

Memory Map for M55s:

Memory Region(WM)	Used Size	Region Size	%age used
ITCM	900880 B	2 MB	42.96%
DTCM	1158400 B	1536 KB	73.65%
LPI_DATA	0 GB	3 MB	0.00%
AOD_DATA	0 GB	3840 KB	0.00%
PSRAM_DATA	0 GB	128 MB	0.00%
DDR	958571 B	4 MB	22.85%
QSR_STR	365414 B	512 KB	69.70%
QSR_NOTPAGED	19184 B	512 KB	3.66%
IDT_LIST	0 GB	2 KB	0.00%

Memory Region(AM)	Used Size	Region Size	%age used
ITCM	915540 B	1280 KB	69.85%
DTCM	770072 B	768 KB	97.92%
LPI_DATA	0 GB	3 MB	0.00%
AOD_DATA	3839 KB	3840 KB	99.97%
PSRAM_DATA	44 MB	128 MB	34.38%
DDR	968507 B	6 MB	15.39%
QSR_STR	103 B	512 KB	0.02%
QSR_NOTPAGED	16 B	512 KB	0.00%
IDT_LIST	0 GB	2 KB	0.00%

6. New Features

Post-CS release for SW6100.LAW.1.0.

New features enabled in this release:

Tools:

- Qprofiler

Connectivity:

- GATT Offload

Display:

- LPI always controls the brightness even when AP SOC is active

7. Limitations

This chapter lists the known issues and limitations reported for this release.

Know Issues

Note: Following are the issues information for this release.

New Issues and Unresolved Issues

CR Number	Title	Description
4354171	[SW6100.LAW.1.0][Audio] ADSP SSR latency is too high(>5s) during Audio/Voice use case	Description: Playback Recovery Delay After ADSP SSR. Steps to reproduce:Initiate audio playback on the DUT.Trigger an ADSP SSR.Observe the time taken for playback to resume after the SSR event. Observed Results:Playback resumes only after a delay of more than 5 seconds post ADSP SSR.The delay is consistent and reproducible. Expected Results:Playback should recover promptly after ADSP SSR, ideally less than 2 seconds.
4368883	[SW6100.LAW.1.0][Display]Panel does not recover after triggering ESD	Description: Panel does not recover after triggering ESD Steps to reproduce: 1. echo 1 /d/*dsi*/esd_trigger 2.Observe the panel status. Observed Results: Panel does not recover Expected Results: Panel shows no display and then recover
4383332	[SW6100.LAW.1.0]HFP AG Call audio issue after SCO reconnection	Setup: ====== DUT- SW6100[MEP1] Role -> HFP AG RD-JBL Flip 4 Use-Case: ========= 1.Pair and connect RD from DUT. 2. Make MO call from DUT and accept call from 3rd party 3. Change call audio path to Speaker 4. Change call audio path to RD Follow the above steps until the issue is observed Observed Results: =============== After Step-2 -> Both Tx and Rx call audio present After step-3 -> No TX and RX call audio After Step 4-> 3rd party call audio on BT headset is glitchy, DUT call audio is heard properly on 3rd party Expected results: ================ After all the steps both TX and RX audio should be present

4387330	[SW6100.LAW.1.0][Audio]Device fails to enter LPI when playback is triggered after touch (low latency) tone completed	Description: Device fails to enter LPI when playback is triggered via Music app play button Steps to reproduce: 1. Connect BT headset to Device 2. check if BT A2DP playback in LPI: adb root adb remount adb shell mount -t debugfs debugfs /sys/kernel/debug In adb shell copy paste this while true do echo "Looping infinitely..." sleep 1 cat /sys/kernel/debug/qcom_stats/adsp_island done 3.playback any clip by using Music app play button 4.check the LPI Stats Count during playback Observed Results: After playback starts, the LPI Stats Count quickly becomes frozen. Expected Results: During the entire playback process, the LPI Stats Count continues to increase.Additional Notes: When touch tone is disabled in Settings, playback via the Music app play button is able to enter LPI.
4458470	[SW6100.LAW.1.0][Audio][Regression]Device LPI Entry Is Unstable During BT A2DP Playback	Description: Device LPI Entry Is Unstable During BT A2DP Playback Steps to reproduce: 1. Connect a Bluetooth headset to the device. 2. Check whether the device enters LPI during BT A2DP playback: adb root adb remount adb shell mount -t debugfs debugfs /sys/kernel/debug In adb shell copy paste this while true do echo "Looping infinitely..." sleep 1 cat /sys/kernel/debug/qcom_stats/adsp_island done 3.Play any audio clip. 4.Observe the LPI stats during playback. Observed Results: The LPI stats count remains at 0 during the entire playback process. Expected Results: The LPI stats count should continue to increase during the entire playback process, and the device should remain in LPI
4461951	[SW6100.LAW.1.0][Display]Occasionally, several regions on the DSI panel exhibit flashing light	Description: Occasionally, several regions on the DSI panel exhibit flashing light. Steps to reproduce: 1. reboot device for several times Observed Results: sometimes, several regions on the DSI panel exhibit flashing light Expected Results: no flashing light observed.

4476355	[ASPEN] [GATT_Offload][WDP]: Server TX bulk transfer fails on repeated iterations	Description: Bulk TX transfer from DUT as GATT server fails on multiple iterations when performed with varying pkt_cnt Steps to reproduce: 1. Register a GATT service with single service & characteristic 2. Once services added, perform LE ACL, where DUT advertises and remote connects to DUT via NRF_connect. Update MTU on remote. 3. Now offload these services with GATT server endpoint ID. 4. After successful offload, received attribute and session ID on LPAI shell. 5. Now perform Server TX Bulk transfer from DUT:bluetooth gatt gattServer_tx_bulk_transferUsage : sessionId attrHandle pkt_cnt val_len Observed Results: 1. When we performed Server Bulk Transfer from DUT with varying <pkt_cnt> for every iteration, we observed Transfer got stuck in between. Post this transfer failure, Un-offload of the current session from HLOS/LPAI fails. Failing to observe prints like session unoffloaded / attribute released prints on HLOS & LPAI Re-offload of this service again also fails with the error "Offload Char failed! status: 144" 2. On build Meta177 :(\\stellar\nsid-blr-02\SW6100.LAW.1.0-00177-STD.GKI.INT-1 ,) with the same scenario, we observed a crash aswell. DUT crashed post TX got stuck :[DROIDBUG-15043108] BDC - [SW6100.LAW.1.0-00177-STD.GKI.INT-1] - LPAI_AM - lpai_bt_gatt_managers.cpp:469 invalid span ptr - Qualcomm JIRA DC2 Expected Results: DUT as gatt server, TX Bulk data transfer should be successful. Repro rate : 3/3 times
4478922	[ASPEN] [GATT_Offload] [WDP]: LPAI Not Notified on GATT Service Removal After Offload	Description: DUT as GATT server, Service Removal After Offload Not Informed to LPAI; Notifications Return Status=0 Steps to reproduce: 1. Register a GATT service - single service & 2 characteristic 2. Once services added, perform LE ACL, where DUT advertises and remote connects to DUT via NRF_connect. Update MTU on remote. 3. Now offload these services with GATT server endpoint ID. 4. After successful offload, received attribute and session ID on LPAI shell. 5. Now perform GATT Server operations from DUT, works fine 6. Now perform Remove Service from HLOS app menu:GATT Server -- RemoveService ServiceUuid: Observed Results: 1. When service is removed on HLOS app menu, we see service removed successfully. RemoveService ServiceUuid:0000a000-0000-1000-8000-00805f9b34fb service removed sucessfully but LPAI is not notified about this change and notifications on the same session return status=0. Expected Results: 1. When service is removed at HLOS, LPAI needs to be notified, and session/attributes needs to be released. 2. Also, notifications on the same session should return an error and not status=0. Repro rate : Seen always

4479706	[SW6100.LAW.1.0] After CIS creation TX voice_audio from DUT not heard on thirdâ€‘party.	Setup: ====== DUT- SW6100[RDL] RD- Samsung Buds2 pro. Use-Case: ========= 1. Connect to DUT from RD. 2. Make MO/MT call on DUT. 3. Once call is active, Disconnect DUT from RD. [still call will be active between DUT and 3rd party device] 4. reconnect to DUT from RD. and check the call_audio presence. Observed Results: =============== After step-4, RX audio able to here on RD, but TX audio from DUT not heard on third?party. Expected results: ================ After step-4, both TX and RX audio should be present [full duplex audio] Analysis =========== CIS creation was successful, and RX audio was audible on the RD; however, TX audio from the DUT was not received on the third?party device.
4482694	[SW6100.LAW.1.0] [Regression] Sports sensor start button icon does not update and function does not work in offload mode (50%).	Description: Sports sensor start button icon does not update and function does not work in offload mode (50%). Steps to reproduce: 1. Enter the tracker after timeout 2. Switch to sports sensor and frequently click the start button Observed Results: Sports sensor start button icon does not update and function does not work in offload mode Expected Results: The start button takes effect after clicking
4484497	[SW6100.LAW.1.0] Observed SCO disconnection after switching the call audio from LC3 to MSBC.	Setup: ====== DUT- SW6100[RDL] RD1- Beates speaker. [MSBC] RD2- Samsung Buds2 pro. [LC3] Use-Case: ========= 1. Connect to DUT from RD1. 2. Make a MO call. Check for voice on RD1. 3. Pair and connect Voice RD2 to DUT. call audio should rout to RD2. 4. Now change active device from Setting UI of DUT. [from RD2 to RD1] Observed Results: =============== After step-4, Observed SCO disconnection and call audio over DUT speaker. Expected results: ================ After step-3, SCO should not disconnect and call audio should be on RD1 speaker.
4487422	[SW6100.LAW.1.0]-After switching between Bluetooth playback and speaker playback, probabilistically meet audio playback with no sound.(6/10)	Test Steps: 1. connect BT 2. Audio playback via BT.ï¼ˆA2DPï¼‰ 3. disconnect BT, switch to speaker playback 4.Repeat above steps for 10 times. Except result: Audio playback normally Test Result: After switching between Bluetooth playback and speaker playback, there is a probabilistic issue where audio plays with no sound. The approximate occurrence rate is 60%

4489273	[SW6100.LAW.1.0][Voice]Bluetooth headset does not play the wake-up success tone when "heysnapdragon" is used to wake SVA in a case without lab parameters(2/10)	Steps to reproduce: 1. Run sva SVA Command with lab data disabled : adb shell am instrument -e class com.qualcomm.aptmm_soundtrigger.SoundTriggerTestBase#testSoundTrigger -e soundModels sm6_HeySnapdragon.PDK6XS_1st_210324_ENPUv6_eAIV5.6.uim -e enableUserVerification0 com.qualcomm.aptmm_soundtrigger/.SoundTriggerTestRunner 2. say keyword "heysnapdragon" you will surely hear a "drip" sound coming from the speaker. 3. BT connects to devices 4. say keyword "heysnapdragon" you will surely hear a "drip" sound coming from BT. but no sound. Observed Results: SVA cannot be woken up by the wak
4485476	[SW6100.LAW.1.0][CNPDT][item_66] - [SW6100.LAW.1.0.r1-00184-STD.INT-1] - pmo_core_enable_wow_in_fw: Failed to receive WoW Enable Ack from FW	Repo in Stability test
4490350	[SW6100.LAW.1.0][CNPDT][item_71] - [SW6100.LAW.1.0.r1-00188-STD.INT-1] - WLAN Crash - Kernel panic - not syncing: subsys-restart: Resetting the SoC wlan crashed	Repo in Stability test

Resolved

- CR 4387355 [Audio] both playback mute while active and ambient playback concurrency
- CR 4384394 [Offload] Start/pause sports sensor in ambient mode with corresponding icon remains unchanged
- CR 4414693 [Offload] Display freeze in ambient mode with ALS enabled
- CR 4400799 [Stability] Monkey test triggers device crash: WLAN_HOST Crash - WLAN Panic @ __dsc_psoc_assert_trans_protected:288
- CR 4411663 [WLAN+BT COEX][STA+SAP+BT][SCC - CONC] : Observing crashes while executing STA+SAP+BT CONC scenarios
- CR 4335068 [BT] BLE LC3 voice call BT_MIC is mute
- CR 4393528 [BT] BT mic mute (1/10)during consecutive BT/Speaker switching headset with aptX-adaptive
- CR 4339728 [LPI] Device becomes unresponsive after executing few usecases with LPI SSR

Limitations

- Part of the devices panel have display intermittent freeze issue, display software team are working with module maker on this issue
- When WLAN is enabled, the system clock reference (XO) remains continuously active. In this configuration, BT and Apps clocks stay well aligned, and time drift remains negligible. When WLAN is disabled and Bluetooth enters sleep, the XO clock is periodically gated to save power. In this low?power mode, continuous clock correlation between BT and Apps cannot be maintained, and time drift naturally accumulates over longer durations. We evaluated multiple clock calibration (XO trim) settings. While some values(0x1A) improve short?term accuracy, clock trim alone cannot prevent long?term drift when the reference clock is not always active. **Periodic (10s) BT Apps time re?synchronization** is recommended. The synchronization interval can be **configured based on customer requirements**, allowing a balance between time accuracy and power consumption.

8. Test Reports

WDP

Test Area		Sub-area	Test Case Name	Procedure	Expect Result	Test Result
Platform	Boot	Boot	EDL download	1. Flash Meta with QFIL(cross test with Fastboot download)	1. Meta flash success, device boot up into UI success	Pass
		Boot	Fastboot download	1. Flash APPS image in fastboot mode(cross test with EDL download, no need test everytime) 2. Run: fastboot reboot	1. Download success	Pass
		Boot	Meta info check	1. cat /vendor/firmware_mnt/verinfo/ver_info.txt	1. Meta info should match	Pass
		Boot	Kernel version check	1. adb shell uname -a	1. Linux Kernel version should be 6.12	Pass
		Boot	Android version check	1. Run: adb root 2. Run: getprop ro.build.version.release	1.Should be "16" (Android W)	Pass
		Boot	Sub-system state	1. Power on device 2. Read Sub-system list, run: # cat /sys/class/remoteproc/*/name 3. Read Sub-system status, run: # cat /sys/class/remoteproc/*/state 4. Check ADSP damon # ps -e | grep rpc	vienna64:/ # cat /sys/class/remoteproc/remoteproc*/name 3000000.remoteproc-adsp 4080000.remoteproc-mss 32300000.remoteproc-cdsp vienna64:/ # cat /sys/class/remoteproc/remoteproc*/state running running running	Pass
		Boot	M55 uart check	1. Power on device, cmd input: adb wait-for-device root && adb shell "vendor/bin/lpi_utility_test 1 1" 2. Connect UART 3. Select second Port, bitrate 4000000 4. Input "help" and check result 5. Input "build_version swm" to check swm version 6. Input "build_version awm" to check awm version	1. M55 UART is on 2. M55 UART cmd can work 3. M55 swm awm version should match	Pass
		Boot	Themisto boot check	1.Power on device 2.Confirm Themisto is running by: cat /sys/class/coproc/smc_state	1.Themisto is running	Pass
		Boot	Load QCN	1. Start QPST and open "software download" window 2. Choose "Restore" and specify the location of QCN 3. Start the QCN Restore 4. Wait "Memory Restore Completed" untill device will auto restart	1. QCN can flash to "Completed" 2. On normal build, will auto-reboot.	Pass
		Boot	UART console login	1. Open Serial COM tool then connect COM Port on debug board 2. Power on the DUT 3. Check if have log output in serial	1. Serial log output success	Pass
		Boot	adb reboot edl	1. Power on device 2. Run: adb reboot edl	1. Device should enter into 9008 mode	Pass
		Boot	adb reboot bootloader	1. Run: adb reboot bootloader 2. Run: fastboot devices	1. Can enter fastboot mode	Pass
		Boot	fastboot reboot	1. Run: adb reboot bootloader 2. Run: fastboot devices 3. Run: fastboot reboot	1. Can enter fastboot mode and reboot	Pass
		Boot	Factory Reset check	1. adb root 2. adb reboot recovery 3. wipe data/factory reset	1. Factory reset success	Pass
		Boot	adb reboot	1. Power on device 2. Run: adb reboot 3. Repeat 1~2 for 5 times	Reboot success	Pass
		Boot	reboot in shell	1. Power on device 2. Run: adb shell 3. Run: reboot 4. Repeat 1~3 for 5 times	Reboot success	Pass
		Boot	adb shell reboot	1. Power on device 2. Run: adb shell reboot 4. Repeat 1~3 for 5 times	Reboot success	Pass
		Boot	adb remount	1. adb root 2. adb remount 3. adb reboot (to disable dm-verity) 4. adb root 5. adb remount	1. Dm-verity disable success 2. adb remount success	Pass
	Peripherals	Peripherals	Aspen UART	1. Open Serial COM tool then connect COM Port on debug board 2. Power on the DUT	Show log information normal	Pass
		Peripherals	LPI UART	1. Open Serial COM tool then connect COM Port on debug board 2. Power on the DUT	Show log information normal	Pass
		Peripherals	Enter fastboot mode with Volume-	1. Long press Volume- button 2. Connect power cable and USB to power on device 3. Release the Volume- button	Device should enter into fastboot	Pass
		Peripherals	Power key to power on	1. Connect Power only 2. Press the Power button for 3s	Device should boot	Pass
		Peripherals	Power key to enter dump mode	1. Power on the device 2. Long press the Power button for 15s then release the button	1. Device should enter into dump mode 2. QPST should collect dump files	Pass
	Linux Kernel	Dump	Full rampdump collection and parse	1. adb shell, run: # echo c > /proc/sysrq-trigger 2. Open QPST, download dump log, don't use PCATApp 3. Dump log parsing with CrashScope	1. Dump mode enter success 2. Dump log download success && auto-reboot after finish download 3. Dump log parse success (at least no apps mismatch)	Pass
		Dump	Dual M55 ramdump collection and parse	1. adb shell, run: # echo c > /proc/sysrq-trigger 2. Open QPST, download dump log, don't use PCATApp 3. Dump log parsing with CrashScope 4. Click CrashScope.Report.csr, check M55 dump under LPAI_SWM and LPAI_AWM	1. After dump parse, can see LPAI_SWM and LPAI_AWM logs	Pass
		Dump	Themisto ramdump collection	1. adb shell, run: # echo c > /proc/sysrq-trigger 2. Open QPST, download dump log, don't use PCATApp 3. Check if have Themisto dump logs	1. Themisto dump logs exist with seven SMC binaries: SMC_BT.bin SMC_PERI.bin SMC_SMC.bin SMC_TME.bin SMC_UWB.bin SMC_WLAN_MM.bin SMC_WLAN_PBL.bin	Pass
		Chipset	CPU Info and idle status	1. Run Catch_cpufreq_ftrace.bat script from \\10.64.29.233\Dropbox\2_Wearable\9_SW5100.LAW.2.0\1_Testlimitation\Catch_cpufreq_trace to catch cpufreq ftrace 2. Open below web and open ftrace file on it to get CPU cur frequency: https://ui.perfetto.dev/, use W,A,S,D to control trace view 3. Get CPU online cores # cat /sys/devices/system/cpu/cpu*/online 4. Get CPU max/min frequency # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_max_freq # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_min_freq # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_cur_freq 5. Get CPU governor # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor 6. Or run Chpiset_test.bat script from \\10.64.29.54\Dropbox\1_Robotic\15_QCS8550.LE.1.0\1_TestLimitation\Platform	1. CPU parameters should match design 2. CPU idle freq should be min level	Pass
		Chipset	GPU Info and idle status	1. Run Chpiset_test.bat script \\10.64.29.54\Dropbox\1_Robotic\15_QCS8550.LE.1.0\1_TestLimitation\Platform 2. Get GPU max/min/cur frequency # cat /sys/class/kgsl/kgsl-3d0/devfreq/max_freq # cat /sys/class/kgsl/kgsl-3d0/devfreq/min_freq # cat /sys/class/kgsl/kgsl-3d0/devfreq/cur_freq	1. GPU parameters should match design 2. GPU idle freq should be min level	Pass
		Chipset	DDR Info and idle status	1. Get DDR idle frequency Powe on device. Wait 5 minutes Check available_frequencies? run: # mount -t debugfs none /sys/kernel/debug # cat /sys/devices/system/cpu/bus_dcvs/DDR/available_frequencies | awk 'END{for(i=1;i<=NF;i++) print int($i/1000)}' Check current frequency? run: # cat /sys/kernel/debug/clk/measure_only_mccc_clk/clk_measure 2. Get DDR max frequency # echo 4224000 > /sys/devices/system/cpu/bus_dcvs/DDR/boost_freq # cat /sys/kernel/debug/clk/measure_only_mccc_clk/clk_measure	1. DDR parameters should match design 2. DDR idle freq should be min level	Pass
	Linux Security	Verity	Enable Verity	1. Run: adb root 2. Run: adb enable-verity	1. verity can be enabled success	Pass
		Verity	Disable verity	1. Run: adb root 2. Run: adb disable-verity 3. Run: adb reboot (for settings to take effect) 4. Run: adb root 5. Run: adb disable-verity	1. verity can be disabled success	Pass
		SELinux	SELinux status	1. Get current SELinux status # getenforce	1. Default SELinux status should be Enforcing	Pass
		SELinux	SELinux config	1. Switch from Enforcing to Permissive # setenforce 0 # getenforce 2. Switch from Permissive to Enforcing # setenforce 1 # getenforce	1. SELinux status config to Permissive success 2. SELinux status config to Enforcing success	Pass
	CPU DCVS (Gold)	WALT Governor	With WALT Governor, efreq and freq of Gold Plus CLKDOM should be FMAX while load is running	1. Run:# echo walt > /sys/devices/system/cpu/cpufreq/policy4/scaling_governor 2. Check governor:# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor 2. Check max freq:# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_max_freq 4. Run cpu loading task on all cpu cores adb push \\10.64.29.233\Dropbox\1_Robotic\26_CQ7790M.LE.1.5_Eliza\1_TestLimitation\stress-ng /data adb shell cd /data chmod 777 stress-ng ./stress-ng --cpu 5 --cpu-load 90 5. Check current cpu freq:# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_cur_freq	1. Gold CPU walt governor success 2. With WALT Governor, cur_freq of Silver cluster should bigger than idle state while load is running	Pass
		Performance Governor	With Performance Governor, freq of Gold Plus CLKDOM should be FMAX irrespective of the efreq of the load	1. Run:# echo performance > /sys/devices/system/cpu/cpufreq/policy4/scaling_governor 2. Check governor:# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor 3. Check max freq:# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_max_freq 4. Check current cpu freq:# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_cur_freq	1. Gold CPU performance governor success 2. With Performance Governor, cur_freq of Silver cluster should always be max freq	Pass
		Powersave Governor	With Powersave Governor, freq of Gold Plus CLKDOM should be FMIN irrespective of the efreq of the load	1. Run:# echo powersave > /sys/devices/system/cpu/cpufreq/policy4/scaling_governor 2. Check governor:# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor 3. Check max freq:# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_max_freq 4. Check current cpu freq:# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_cur_freq	1. Gold CPU powersave governor success 2. With powersave Governor, cur_freq of Silver cluster should always be min freq	Pass
		Check frequency range	Check frequency range	Check freq range :# cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_available_frequencies	1. scaling_available_frequencies is normal	Pass
Security		Security	QSEECOM	1. Run: # qseecom_sample_client v smplap64 1 5	1. QSEECOM run pass	Pass

<table>
<tr><th></th><th></th><th></th><th></th><th></th><th></th><th></th></tr>
<tr><td>Sub-system</td><td></td><td>Sub-system state</td><td>Sub-system state</td><td>1. Power on device
2. Read Sub-system list, run:
cat /sys/class/remoteproc/*/name
3. Read Sub-system status, run:
cat /sys/class/remoteproc/*/state
4. Check ADSP damon
ps -e | grep rpc</td><td>1. Sub-system status correct
2. adsprpc cdsproc damon running</td><td>Pass</td></tr>
<tr><td></td><td></td><td>SSR test on ADSP</td><td>SSR test on ADSP</td><td>adb root
adb remount
adb shell
mount -t debugfs none /sys/kernel/debug
echo enabled > /sys/class/remoteproc/remoteproc0/recovery
Crash inject command:
ADSP QXDM command: send_data 75 37 03 48 02

1. Check SSR count :
cat /sys/class/remoteproc/remoteproc0/device/txn_id
Check SSR log :
dmesg | grep "remoteproc"</td><td>Can see adsp subsystem restart completed info in demsg log</td><td>Pass</td></tr>
<tr><td></td><td></td><td>SSR test on CDSP</td><td>SSR test on CDSP</td><td>1. Power on device
2. Connect to QXDM
3. Run below command to get the subsystem ID
cat /sys/class/remoteproc/remoteproc*/name
4. Enable SSR
echo enabled > /sys/class/remoteproc/remoteproc2/recovery
5. Send SSR command
send_data 75 37 03 96 00
6. Check SSR count :
cat /sys/class/remoteproc/remoteproc2/device/txn_id
Check SSR log :
dmesg | grep "cdsp is"</td><td>Can see cdsp subsystem restart completed info in demsg log</td><td>Pass</td></tr>
<tr><td></td><td></td><td>SSR test on MODEM</td><td>SSR test on MODEM</td><td>1. Power on device
2. Connect to QXDM
3. Run below command to get the subsystem ID
cat /sys/class/remoteproc/remoteproc1/name
4. Enable SSR
echo enabled > /sys/class/remoteproc/remoteproc1/recovery
5. In QXDM , send SSR command
send_data 75 37 03 00 00
6. Check SSR count :
cat /sys/class/remoteproc/remoteproc1/device/txn_id
Check SSR log :
dmesg | grep "mss is" and search keywords:"mss is"?dmesg | grep "mpss subsystem is"?</td><td>Can see modem subsystem restart completed info in demsg log</td><td>Pass</td></tr>
<tr><td></td><td></td><td>SSR test on M55</td><td>SSR test on M55</td><td>adb root
adb remount
adb shell
mount -t debugfs none /sys/kernel/debug
echo 1 > /sys/module/qcom_ramdump/parameters/enable_dump_collection
echo enabled > /sys/class/remoteproc/remoteproc0/recovery
echo enabled > /d/remoteproc/remoteproc0/coredump
echo inline > /d/remoteproc/remoteproc0/coredump
/vendor/bin/subsystem_ramdump &
Crash inject command:
ADSP QXDM command: send_data 75 37 03 48 02
Pull SSR dump from Device instead of PCAT
adb pull /data/vendor/ramdump <destination path></td><td>1.adsp.elf file generated.</td><td>Pass</td></tr>
<tr><td></td><td></td><td>SSR test on Themisto</td><td>SSR test on Themisto</td><td>1. Trigger Themisto SSR
cat /sys/class/coproc/smc_state
echo enabled > /sys/class/coproc/recovery
echo enabled > /sys/class/coproc/coredump //command to crash SMC
/vendor/bin/smcpblupdater -c1
cat /sys/class/coproc/smc_state
2. Collect dump
adb root
adb remount
adb pull /data/vendor/ramdump</td><td>1.SMC_-2025-11-22-16-46-24.452.bin file generated.</td><td>Pass</td></tr>
<tr><td>FOTA</td><td></td><td>FOTA</td><td>Full-OTA upgrade</td><td>1. Using update package and ota_call.py to get update command:
$ python ota_call.py update.zip for hash output.
2. adb push msm8998-target_files-eng.zip /data/ota_package/update.zip
3. adb root (Don't excute adb remount!!or do adb enable-verity)
4. adb shell
5. Run the upgrade command which get from ota_call.py on step1
eg:
#update_engine_client --update --follow --payload=file:///data/ota_package/update.zip --offset=4682 --size=1110308359 --headers="FILE_HASH=UPga42iKPRSw/+hWMbb9m8o2Ocb7Y7Aq6wh2Og/33UM=
FILE_SIZE=1110308359
METADATA_HASH=PJ3x1bIkf8ajaGPz3huFN6+RezsCFZp4jzkS/6r6HxU=
METADATA_SIZE=88800
"

7. In Parallel collect update engine logs.
8. adb logcat | grep -I update_engine
9. After update, reboot request is thrown in logs capture command prompt.
10. Reboot the device.
11. Device will boot from inactive slot with updated build.
12.Check boot slot after OTA
adb shell bootctl get-current-slot</td><td>1. FOTA update should success</td><td>Pass</td></tr>
<tr><td></td><td></td><td></td><td>Ambient mode enter and exit after OTA</td><td>1. Go to Settings -> Display, set AON to ON
2. Set device in Idle: APM off, WLAN connect to AP, USB dis-connected, BT paired with Headset
3. Set time-out as 1min (by default)
4. Back to Watch Face
5. Connect USB/adb
6. Wait to time-out and offload happens
7. Check in logs if status match
8. Click Power key
9. Check if mode exit success
1) UI offload back to Active
2) Have exit info in log
10. Check BT connection during mode exit
11. Check status after mode exit
1) APM off and Network register success
2) WLAN connect to AP</td><td>1. APSS Status should be awake
2. Display offload should work
3. Mode status should be Ambient in logs
4. Mode exit success
5. BT always connected during mode exit
6. Status correct after mode exit</td><td>Pass</td></tr>
<tr><td>MM</td><td>Camera</td><td>Front Camera Preview</td><td>Front _Camera Preview srMax</td><td>1. Open the Camera App and Disable ZSL
2. Take Preview with Selected Resolution, Duration 1min</td><td>1.Camera Preview should be good.
2.On changing the resolution and doing Preview, No ANR or any corruptions should be observed in Preview and Resolution is correct.</td><td>Pass</td></tr>
<tr><td></td><td></td><td>Front Camera Snapshot</td><td>Front_Camera ZSL Snapshot srMax12MP</td><td>1. Open the Camera App with zsl on
2. Of Max Resolutions, Take Preview & Snapshot for 3times
3. Pull out the snapshot to PC to check</td><td>1. Camera Preview should be good.
2. On changing the Resolutions and Taking Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview as well as Snapshot. and the resolution is correct
3. The JPEG snapshot and preview should be of the same object of interest. Colors should be reproduced accurately.</td><td>Pass</td></tr>
<tr><td></td><td></td><td>Front Camera Live Snapshot</td><td>Front_Camcorder live snapshot_Video_h264_Max_LS_1080p60fps</td><td>1. Open the Camera App.
2. Of required Resolutions, Take live Snapshot during video recording for 3times
3. Pull out to PC to check</td><td>1. Camera Preview should be good.
2. On changing the Resolutions and Taking the live Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview, live snapshot as well as video clip.
3. the resolution of video and the live snapshot is correct
4. The live snapshot and video should be of the same object of interest. Colors should be reproduced accurately.</td><td>Pass</td></tr>
<tr><td></td><td></td><td>Front Camera Video_Encode</td><td>Front_Camcorder Recording-Max Resulution/FPS</td><td>1. Launch Camera
2. Switch to camcorder
3. Select settings
4. Select Max resulion 1080p/60fps video quality
5. Start record
6. Click Paust/Start during recording 5 times
7. Stop record</td><td>1. Can video recording
2. File save success and video size and fps is match
3. Can pause/start during playing
4. video playback normally
5. Can hear sound normally</td><td>Pass</td></tr>
<tr><td></td><td></td><td>Back Camera Preview</td><td>Back_Camera Preview srMax</td><td>1. Open the Camera App and Disable ZSL
2. Take Preview with Selected Resolution, Duration 1min</td><td>1. Camera Preview should be good.
2. On changing the resolution and doing Preview, No ANR or any corruptions should be observed in Preview and Resolution is correct.</td><td>Pass</td></tr>
<tr><td></td><td></td><td>Back Camera Snapshot</td><td>Back_Camera ZSL Snapshot srMax12MP</td><td>1. Open the Camera App with zsl on
2. Of Max Resolutions, Take Preview & Snapshot for 3times
3. pull out the snapshot to PC to check</td><td>1. Camera Preview should be good.
2. On changing the Resolutions and Taking Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview as well as Snapshot. and the resolution is correct

3.The JPEG snapshot and preview should be of the same object of interest. Colors should be reproduced accurately.</td><td>Pass</td></tr>
<tr><td></td><td></td><td>Back Camera Live Snapshot</td><td>Back_Camcorder live snapshot_Video_h265_Max_LS_1080p60fps</td><td>1. Open the Camera App.
2. Of required Resolutions, Take live Snapshot during video recording for 3times
3. Pull out to PC to check</td><td>1. Camera Preview should be good.
2. On changing the Resolutions and Taking the live Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview, live snapshot as well as video clip.
3. the resolution of video and the live snapshot is correct

The live snapshot and video should be of the same object of interest. Colors should be reproduced accurately.</td><td>Pass</td></tr>
<tr><td></td><td></td><td>Back Camera Video_Encode</td><td>Back_Camcorder Recording-Max Resulution/FPS</td><td>1. Launch Camera
2. Switch to camcorder
3. Select settings
4. Select Max resulion1080p/60fps video quality
5. Start record
6. Click Paust/Start during recording 5 times
7. Stop record</td><td>1. Can video recording
2. File save success and video size and fps is match
3. Can pause/start during playing
4. video playback normally</td><td>Pass</td></tr>
<tr><td></td><td></td><td>FD</td><td>Front_Camera_Video_Recording_720P30_H264_Duration30s</td><td>1. SD camera app in installed in the devices
2. FD UI is enabled
1. Open Camera app and switch to front camera
2. Switch mode from photo to video
3. Go to settings to enabled FD
4. Towards face and then Set the same video parameters with case described
5. Press start key to video record
6. Wait for duration time, check video recording and generate file</td><td>1. towards face, we can see face ROI in preview
2. video recording will stop automatically at the end of the set time
2. Playback video file by Snapdragon Gallery app, video file duration, fps, quality, resolution are the same with primary set
3. App not find ANR,Crash, Hung</td><td>Pass</td></tr>
<tr><td></td><td></td><td>MCTF</td><td>Front_Camera_Video_Recording_720P30_H264_Duration30s</td><td>1. SD camera app in installed in the devices
2.MCTF is enabled
1. Open Camera app and switch to front camera
2. Switch mode from photo to video
3. Enable MCTF
4. Set the same video parameters with case described
5. in the low light condition, press start key to video record
6. Wait for duration time, check video recording and generate file
7.Disable MCTF and repeat the same steps</td><td>1. video recording will stop automatically at the end of the set time
2. Playback video file by Snapdragon Gallery app, video file duration, fps, quality, resolution are the same with primary set
3. App not find ANR,Crash, Hung
4. Less noise in MCTF enabled video clip compared with MCTF disabled video clip</td><td>Pass</td></tr>
<tr><td></td><td></td><td>Video HFR - 120FPS</td><td>Front_Camera_Video_Recording_720p120fps_H264_Duration10mins</td><td>1. SD camera app in installed in the devices
2. HFR option enabled in UI
1. Open Camera app
2. Switch mode from photo to HFR
3. HFR record parameter set the same with case name describe
4. Press start key to HFR record
5. Wait for duration time, check video recording and generate file</td><td>1. HFR recording will stop automatically at the end of the set time
2. Playback HFR file by Snapdragon Gallery app, video file duration, encode format, fps, quality, resolution are the same with primary set
3. App not find ANR,Crash, Hung</td><td>Pass</td></tr>
<tr><td></td><td></td><td>Video EIS</td><td>Front_Camera_VideoRecording_720P30_H264_DurationNolimit_EIS on</td><td>1. SD camera app in installed in the devices
2. image Stabilization setting is functional in SD camera to enable EIS
3. Video EIS setting is functional in SD camera to enable EIS v3
4. Dual TFE based 1080p + 50% margin for video output from camera is enabled

1. Open Camera app?
2. Switch mode from photo to video
3. Record parameter set the same with case name describe, enable EIS by set image Stabilization on, keep device on repeat Slight shaking, press start key to video record, wait for duration time.
4. Disable EIS by set image Stabilization off, , keep device on repeat Slight shaking, press start key to video record, wait for duration time.
5.Pull out and check video recording and generate file by PC</td><td></td><td>Pass</td></tr>
<tr><td></td><td></td><td>Shutter sound</td><td>Camera Shutter sound</td><td>1. Take Preview & Snapshot on Front and Back camera, hear camera shutter sound</td><td>1. Can hear camera shutter sound</td><td>Pass</td></tr>
</table>

	Video	Video_Decode	Video_playback VP9	1. Push VP9 video file /sdcard/Movies 2. Open Gallery 3. Playback	Video Playback normally	Pass
		Video_Decode	Video_playback HEVC(H265)	1. Push H265 video file /sdcard/Movies 2. Open Gallery 3. Playback	Video Playback normally	Pass
		Video_Decode	Video_playback H264	1. Push H264 video file /sdcard/Movies 2. Open Gallery 3. Playback	Video Playback normally	Pass
		Video_Decode	Video_playback Seek/Forward/Back/Volume/Brightness adjust	1. Playback a video 2. Try Seek, just jump to another timestamp 3. Try Forward/Back, scroll right and left for 3times 4. Try adjust Volume, scroll up and down in right aread for 3times 5. Try adjust Brightness, scroll up and down in left aread for 3times 6. Try adjust Volume via HW Volume up button for 3times	Seek/Forward/Back/Volume/Brightness adjust works	Pass
	Audio	Audio_Encode	AE_encode_by_soundrecord_apk	1. Select AMRNB as a recording format. 2. Start recording by soundrecord apk. 3. Stop the recording. 4. Save the recording.	1. Recorded clip should have some content/data. 2. No pops or glitches should be heard in a recording 3. No crashes observed during the recording. 4. Recording time and recorded clip duration should be the same.	Pass
		Audio_Decode	AD_CompressedOffload_001_wav_playback	1. Play wav > 60s using Music app	1. Verify that compressed-offload clip plays with audio heard all the time and no crashes are seen in the logs 2. Verify in logcat for "out_set_parameters" being set to "compress-offload-playback"	Pass
		Audio_Decode	AD_CompressedOffload_001_mp3_playback	1. Play mp3 > 60s using Music app	1. Verify that compressed-offload clip plays with audio heard all the time and no crashes are seen in the logs 2. Verify in logcat for "out_set_parameters" being set to "compress-offload-playback"	Pass
		Audio_Decode	AD_CompressedOffload_001_aac_playback	1. Play aac > 60 s using Music app	1. Verify that compressed-offload clip plays with audio heard all the time and no crashes are seen in the logs 2. Verify in logcat for "out_set_parameters" being set to "compress-offload-playback"	Pass
		Audio_Decode	AD_musicapp_playback_hw_volume_profile	1. Playback audio using Music app 2. Adjust volume from mute to max and back to mute by HW volume buttons 3. Verify if playback volume changed during audio playback	1. No crashes or ANRs 2. No loss in audio or glich/pops. 3. Volume displays an exponential profile, with same levels as PCM. 4. Volume level display moves accordingly. 5. Soft stepping is enabled 6. No saturation at max volumes	Pass
		Speaker_Protection	AD_FBSP_quick_calibration_noplayback	1.adb pull /vendor/etc/audio/sku_vienna64/resourcemanager_vienna.xml <param key="logging_level" value ="7" /> (change 3 to 7) <speaker_protection_enabled>1</speaker_protection_enabled> (change 0 to 1) <quick_cal_time>90</quick_cal_time> (change 90) adb push C:\Users\shafeng\resourcemanager_vienna.xml /vendor/etc/audio/sku_vienna64/resourcemanager_vienna.xml adb reboot wait 90s # adb shell logcat -v threadtime | grep -sn "SpeakerProtection" ? to check log kyewords "SpeakerProtection"? #cd /mnt/vendor/persist/hlos_rfs/shared #ls -l and check keywords" audio.cal"	1. Verify calibration success in logcat log(adb logcat | grep SpeakerProtection) 2. Verify calibration file(audio.cal) generated in file system	Pass
		Speaker_Protection	AD_FBSP_audio_playback_speaker	1. Playback music after setting below steps	1. No audible pops/clicks heard	Pass
		Fluence	VoIP_MO_Call_with_Fluence_enabled_DMEF	1.Run C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\Audio\Fluence\push_voip_dmef_xml.bat 2.install xiaohongshu.apk 3.Open QACT Tool,and connect device 4. make a MO Wechat voice call 5.Check if the QACT tool displays the word 'fluence'	The sound output correct with no noise	Pass
		Fluence	VoIP_MT_Call_with_Fluence_enabled_DMEF	1.Run C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\Audio\Fluence\push_voip_dmef_xml.bat 2.install xiaohongshu.apk 3.Open QACT Tool,and connect device 4. make a MT Wechat voice call 5.Check if the QACT tool displays the word 'fluence'	The sound output correct with no noise	Pass
		Haptics	AD_Haptics_with_Alarm	1. Enable haptics in settings 2. Set Offload app alarm with some time 3. Obeserve alarm with haptics	1. Verify haptics with Offload app alarm	Pass
		Speaker_Protection_ambient	AD_FBSP_audio_ambient_playback_notification_tone_speaker	adb push \\louisiana\SZTestTeam\2025\IOT_Audio_Log\Aspen\FC\LPAI_WM_363-LAW\. /vendor/firmware_mnt/image/ 1.Enable speaker protection and make sure audio.cal file is generated? #cd /mnt/vendor/persist/hlos_rfs/shared #ls -l 2.Open QXDM and connect to Device to collect log File-LoadConfiguration?C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\Speaker_Protection? 3.refer 2.3 to run Notification and touch tone playback in ambient mode Test Cases on UART shell: a. enable The M55 uart shell b.input command in M55 uart shell : #audio_service run notif 1 0.3 4.Save QXDM log File->Save Items 5.Open QCAT to analysis QXDM log. File->Open->File->select QXDM log View->Vocoder Playback->Process check the output in Output folder	1.check if 1586 contain 0xE07/17/18	Pass
		Speaker_Protection_ambient	AD_FBSP_audio_ambient_playback_touch_tone_speaker	adb push \\louisiana\SZTestTeam\2025\IOT_Audio_Log\Aspen\FC\LPAI_WM_363-LAW\. /vendor/firmware_mnt/image/ 1.Enable speaker protection and make sure audio.cal file is generated? #cd /mnt/vendor/persist/hlos_rfs/shared #ls -l 2.Open QXDM and connect to Device to collect log File-LoadConfiguration?C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\Speaker_Protection? 3.refer 2.3 to run Notification and touch tone playback in ambient mode Test Cases on UART shell: a. enable The M55 uart shell b.input command in M55 uart shell : #audio_service run touch 1 0.012 4.Save QXDM log File->Save Items 5.Open QCAT to analysis QXDM log. File->Open->File->select QXDM log View->Vocoder Playback->Process check the output in Output folder 6.check if 1586 contain 0xE07/17/18	1.check if 1586 contain 0xE07/17/18	Pass
		Audio_Decode	audio playback resume post SSR	1.#cat /sys/class/remoteproc/remoteproc*/name #echo enabled > /sys/class/remoteproc/remoteproc0/recovery (Choose the adsp remotepro) 2.Start playback music with pulseaudio 3.In QXDM, give the following command:send_data 75 37 03 48 00 4.Playback/Record will stop and resume. 5.Check SSR count #cat /sys/class/remoteproc/remoteproc0/device/txn_id 6.Check SSR log adb shell dmesg | grep "Q6 is"	After triggering SSR, Playback/Record will stop and resume	Pass
		Audio_Decode	audio record resume post SSR with BT	1.BT connect to headset 2.#cat /sys/class/remoteproc/remoteproc*/name #echo enabled > /sys/class/remoteproc/remoteproc0/recovery (Choose the adsp remotepro) 3.Start record with pulseaudio 4.In QXDM, give the following command:send_data 75 37 03 48 00 5.Playback/Record will stop and resume. 6.Check SSR count #cat /sys/class/remoteproc/remoteproc0/device/txn_id 7.Check SSR log adb shell dmesg | grep "Q6 is"	After triggering SSR, Record stop and BT disconnected, will resume later	Pass
	Display	Display	DI_Brightness	1. Open the Down bar 2. Change the Brightness 3. Check the brightness is changed	Brightness can be changed	Pass
		DI_Scrolling	Check if scrolling is even. Settings -> Scroll -> wait 2 Sec. for fallback	Scrolling down should be free of Janks or stutters.	Scrolling is even	Pass
	Touch	Touch	Single tap	1.Touch to open a app 2.Touch to scroll the app menu	Touch works smoothly	Pass
Sensor		Physical Sensor	Accel	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Place DUT for UP/DOWN/LEFT/RIGHT/FRONT/BACK six different orientation	1. Accel(x, y, z) data should corresponding to the place orientation 2. For example, when place UP, should be (0, 0, 9.8)	Pass
		Physical Sensor	Gyro	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Rotation DUT on YAW/ROLL/PITCH three different orientation	1. Gyro(x, y,z) data change should corresponding to the rotation orientation 2. For example, when rotation on YAW, only Z data change	Pass
		Physical Sensor	Mag	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Place DUT on different orientation	Mag(x, y,z) data should corresponding to the place orientation	Pass
		Physical Sensor	ALS/Light	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Use hand to cover and then left the DUT	Light data should decrease and the increase	Pass
		Physical Sensor	Pressure	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type	Pressure data should be reasonable	Pass
		Physical Sensor	Skin temperature	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Use hand to click Skin temperature sensor	Skin temperature data should have change when click by hand	Pass

Connectivity	Co-work	WLAN/BT Co-work	WLAN+BT enable at same time	1. Open wifi menu 2. Enable bt and pair with headset for audio playback 3. Open wifi menu 4. Enable wifi and connect AP to ping 5. Check the BT/Wlan all open at same time	WLAN+BT co-work success	Pass
	WLAN	STA	STA: Connect to an Ref-AP with Password	1. Connect to an AP which has password setup (WPA2) 2. Enter the password and ensure that connection can be established.	1. Wifi can enable success 2. Wifi ssid will show in list 3. Input password can connect success	Pass
		STA	STA: Browse the web or Ping a Web site	1. Open Browser 2. Input a Web link 3. Or ping a Web site like: www.baidu.com	1. Connect "Hydra" success 2. Can open the browser and view the web page	Pass
		Remember	STA: Remember Network	1. After connecting to a network, select "Remember network" 2. Disable WLAN and then Enable WLAN for 3times 3. Ensure that the network which was saved earlier still shows up.	Auto connect success	Pass
	BT	Pair	BT Device Pair	1. Switch ON BT. 2. Search for BT Devices. 3. Pair to the Devices. 4. Pair and unpair BT 3 times	1. BT enable success 2. Show search BT devices list 3. Can pair the any devices	Pass
		A2DP	Playback over BT Headset	1. Switch on BT 2. Search BT Headset Devices. 3. Bond to an Identified BT Headset Device. (A2DP) 4. Start MP3 Playback and check for Audio Playback over the headset. Check for subjective audio quality. 5. Increase/decrease volume via DUT HW button and BT Volume button during BT Headset Device 6. Repeat steps 5 for 3times via DUT HW button and BT Headset button	1. BT enable success 2. Show search BT devices list 3. Can pair the any devices 4. Audio can playback normally via headset 5. Volume can be adjusted	Pass
		AVRCP	Pause/Start/Next/Last over BT Headset	1. Pair on BT headset and play audio 2. On BT headset, click Pause/Start/Next/Last sound via APP in DUT side for 3times 3. On BT headset, click Pause/Start/Next/Last sound via Headset button for 3times	1. Pause/Start/Next/Last works well via app and BT Headset button control	Pass
		HFP	Voice Call over BT Headset	1. MO/MT Call with BT Headset connected 4. During voice call,Increase/decrease volume via DUT HW button and BT Headset button, Repeat six times each	1.Voice call with BT Headset connected normally 2. Sound recognized normally when swith BT Headset	Pass
		Device Switch	DeviceSwitch_Playback_speaker_BT	1. Playback over BT Headset 2. Swith BT Headset Devices to Speaker via APP in DUT side, 3. Restart audio playback when swith to device. 4. And then swith Speaker to BT Headset via BT Headset button 5. Repeat 1~4 for 5times	1. BT/Speaker switch success via APP and BT Headset button control 2. No delay in voice recognition after swithed	Pass
		Rememer	BT Device Pair remember	1. Disable BT 2. Then enable BT 3. Enable/Disable BT for 3 times 4. Wait 10s to check if BT can auto pair to device	Auto pair success	Pass
	UWB	UWB	UWB Enable	1. Power on device 2. You can enable UWB by UI: Settings--Connected devices--Connection preferences--UWB 3. Or you can trigger enablement of UWB via shell command force-country-code? adb shell "cmd uwb force-country-code enabled US" cat uwb state: adb shell cmd uwb status enable uwb: adb shell cmd uwb enable-uwb	1. UWB is enable	Pass
	NFC	Enable/Disable	NFC enable and Disable	1. Go to Menu -->Settings--> NFC 2. Turn on NFC and disable/enable for 3times	1. NFC enable/disable success	Pass
		NFC tag read	NFC tag read	1. Install NFC tag apk: \\jianwu22\Dropbox\2_Wearable\11_SW6100.LAW.1.0\01.TestLimitation\NFC\NFC Tools_v8.6_apkpure.com 1.apk 2. Turn on NFC and enable it 3. Open NFC tag app, read tag	1. NFC tag can be read	Pass
Telephony		IMS register	Register IMS	1. Inset a SIM card support VoLTE or VoINR	1. When IMS register success, should have HD tag	Pass
		VoLTE	VoLTE_MO answer	1.Camp to VoLTE network and register IMS 2.Make MO Call 3.After MO call get connected Make MT call 4.Device reject the MT call 5.Device answer the MT call	1.Mo call get connected, the first active call should hold.	Pass
		VoLTE	VoLTE_MT anwser	1.Camp to VoLTE network and register IMS 2.Make MT Call 3.After MT call get connected Make MO call	1.devices can reject the MT call, and the first active call should keep . 2.device can answer the MT call , the first call hold.	Pass
		VoNR	VoNR_MO answer	1. Camp to NR network and register IMS 2. Make MO Call	1. When IMS register success, should have HD tag 2. MO call should connect, during the call, network mode should still be 5G 3. Data still works using Ping	Pass
		VoNR	VoNR_MT answer	1. Camp to NR network and register IMS 2. Make MT Call	1. When IMS register success, should have HD tag 2. MO call should connect, during the call, network mode should still be 5G 3. Data still works using Ping	Pass
		SMS VoNR	SMS VoNR	1. Send MO SMS to other devices	1. Mo SMS send from DUT must be receveid .	Pass
		SMS VoNR	SMS VoNR	1. Make MT SMS from other device to DUT	1. DUT should receive MT SMS .	Pass
		Data-Celluar	LTE(4G)_Browse	1.Do browsing	1. Data browsing should happen	Pass
		Data-Celluar	NR(5G)_Browse	1.Do browsing	1. Data browsing should happen	Pass
		Voice-Common	Voice-Common_Volume change during call	1. Make MO Call 2. Change the volume when Call is active	1. Option to change volume should came when call is active . 2. We should able to increase or decrese the volume	Pass
		Emergency call	Emergency call-110/120/119	1. Make MO Emergency call by dailing 110/120/119	1. Call should connect to 110/120/119, check dialed number , and Call timer updates	Pass
		Adversial-APM	Adversial-APM_ON-OFF	1.Put dut in APM, dial any non emergency number 2.Off the APM, Make MO call	1. dial shouldn't work, it should show to dial take dut out of APM 2.Call should get connected.	Pass
Location		GPS	Standalone app tracking	1. Install GPS-test Apk on the target 2. Open the GPS Test App. 3. Test outside, wait 1~2min	1. GPS/beidou/GLONSS satellites all can be found 2.Latitude/Longitude position info can be gived	Pass
Wear Tech		UI	Date/time auto sync by WWAN	1. Register to NR network 2. Check if date/time can auto sync	1. Date/time can auto sync	Pass
			Watch Face and Sports UI switch in tile	1.Enable the UI Swiping by: adb root; adb remount adb shell setprop persist.watchui.quickSettingsEnabled true reboot 2.Left swipe to sports UI from watch face UI Right swipe to watch face UI from sports UI 3. repeat 5 times	1.Switch the UI successfully, the UI content is correct	Pass
			Watch Face switch	1. Go to Face setting, choose digital watch face 2. Switch to analog watch face 3. Switch to digital watch face with sports 4. Switch to analog watch face with sports	Each watch face switch success, and UI display normal	Pass
			Enter Quick settings panel	1.Slide from top to bottom on the watchface 2.Check the Quick settings panel display, include: 1) 5 buttons: WIFI ON/OFF, AirplaneON/OFF, Bluetooth ON/OFF, Settings, Brightness 2) 3 icons: Battery status,Telecommunications signal, Time	Enter Successfully and Display correctly 1.Buttons: Include a pattern in the middle with a circle on the outside Icons: 1) Battery status(display charging icon while charging or display remaining battery level when not charge) 2) Telecom signal icon(display signal status bright or dark, show signal type like LTE,UMTS and so on upper left of the icon 3) Time and date (displays the current hours, minutes, seconds, day of the week, and date)	Pass
			Click the Settings button	1.Click the Settings button to jump to the Setting page	1.Jump to the Settings page successfully	Pass
			Restart	1. Long press power key or go to Settings -> Restart 2. Perform restart and observe the device behavior	Restart function and UI works well	Pass
			Power off	1. Long press power key or go to Settings -> Power Off 2. Perform power off and observe the device behavior	Power off function and UI works well	Pass
			Factory Data Reset	1. Go to Menu -->Settings--> Factory Reset 2. Perform factory data reset and observe the device behavior	1. All user related data should be removed and device should be as it was after first boot up.	Pass
		Offload	Ambient mode enter and exit	1. Go to Settings -> Display, set AON to ON 2. Set device in Idle: APM off, WLAN connect to AP, BT paired with Headset 3. Set time-out as 15sec (by default) 4. Back to Watch Face 5. Connect USB/adb 6. Wait to time-out and offload happens 7. Check if display offload work 8. Check in logs if status match 9. Click Power key 10. Check if mode exit success 1) UI offload back to Active 2) Have exit info in log 11. Check BT connection during mode exit 12. Check status after mode exit 1) APM off and Network register success 2) WLAN connect to AP	1. Display offload should work 3. Mode status should match in logs 4. Mode exit success 5. BT always connected during mode exit 6. Status correct after mode exit	Pass
			AON on under Ambient mode	1. Go to Settings -> Display, set AON to ON 2. Set device in Idle: APM off, WLAN connect to AP, BT paired with Headset 3. Set time-out as 1min (by default) 4. Back to Watch Face 5. Wait to time-out and offload happens	1. Display offload should work and display should be in always-on mode	Pass
			Reboot under ambient mode	1. Enter Ambient mode 2. Reboot DUT under Ambient mode 3. Repeat 5 times	1. Reboot success under Ambient mode	Pass
ML		QNN	QNN on NSP	C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\QNNTest\QNN\qnn-sdk-v2.37.1.25072\Push_QNN.bat adb push lib\aarch64-android\. /vendor/lib/ adb push bin\aarch64-android\. /vendor/bin/ adb push lib\hexagon-v81\unsigned\. /vendor/lib/rfsa/adsp/ adb push \\lab8510\Dropbox\AISW_132328\qnn-test-v2.37.0.25_Test_pack\qnn-test-v2.37.0.250724175447_124859\model_tf2_10_custom_mlperf_deeplabv3_qat\inputs\. /data/inputs adb push \\lab8510\Dropbox\AISW_132328\qnn-test-v2.37.0.25_Test_pack\qnn-test-v2.37.0.250724175447_124859\model_tf2_10_custom_mlperf_deeplabv3_qat\quantized\aarch64-android\tf2_10_custom_mlperf_deeplabv3_qat_quantized.so /vendor/ Executed command: # cd /data/inputs # qnn-net-run --log_level=error --profiling_level=basic --backend=/vendor/lib/libQnnHtp.so --input_list=/data/inputs/input_list.txt --output_dir=/data/inputs/ --model=/vendor/tf2_10_custom_mlperf_deeplabv3_qat_quantized.so Get Inference stats: # qnn-profile-viewer --input_log /data/inputs/qnn-profiling-data.log	Execute Stats (Average): ------------------------ Total Inference Time: --------------------- Graph 0 (qnn_model): NetRun: 14210 us Backend (Number of HVX threads used): 4 count Backend (RPC (execute) time): 13181 us Backend (QNN accelerator (execute) time): 11184 us Backend (Accelerator (execute) time): 10994 us Backend (Accelerator (execute excluding wait) time): 10324 us Backend (QNN (execute) time): 14381 us	Pass

Product	Power	Suspend check with SIM card and APM OFF	1. Insert SIM card and Power on device
			# adb shell "mount -t debugfs none /sys/kernel/debug"
			2. adb root and run below commands
			  # cat /d/suspend_stats
			3. Disconnect USB and wait for time-out or use power key to suspend
			4. After display cut off, wait 30s
			5. Connect USB or use power key to resume
			6. Check if enter suspend success
			  # cat /d/suspend_stats
			7. Repeat 3~6 for 3times
				Suspend enter success	Pass

		Suspend check with SIM card and APM ON	1. Insert SIM card and Power on device, then open Airplane mode
			# adb shell "mount -t debugfs none /sys/kernel/debug"
			2. adb root and run below commands
			  # cat /d/suspend_stats
			3. Disconnect USB and wait for time-out or use power key to suspend
			4. After display cut off, wait 30s
			5. Connect USB or use power key to resume
			6. Check if enter suspend
			  # cat /d/suspend_stats
			7. Repeat 3~6 for 3times
				Suspend enter success	Pass

		Vddmin check with SIM card and APM OFF	1. Insert SIM card and Power on device
			# adb shell "mount -t debugfs none /sys/kernel/debug"
			2. adb root and run below commands
			  # cat /d/suspend_stats
			  # cat /d/qcom_stats/aosd
			  # cat /d/qcom_stats/cxsd
			3. Disconnect USB and wait for time-out or use power key to suspend
			4. After display cut off, wait 30s
			5. Connect USB or use power key to resume
			6. Check if enter suspend/vddmin success
			  # cat /d/suspend_stats
			  # cat /d/qcom_stats/aosd
			  # cat /d/qcom_stats/cxsd
			7. Repeat 3~6 for 3times
				Vddmin enter success	Pass

		Vddmin check with SIM card and APM ON	1. Insert SIM card and Power on device, then open Airplane mode
			# adb shell "mount -t debugfs none /sys/kernel/debug"
			2. adb root and run below commands
			  # cat /d/suspend_stats
			  # cat /sys/kernel/debug/qcom_sleep_stats/vmin
			3. Disconnect USB and wait for time-out or use power key to suspend
			4. After display cut off, wait 30s
			5. Connect USB or use power key to resume
			6. Check if enter suspend/vddmin success
			  # cat /d/suspend_stats
			  # cat /sys/kernel/debug/qcom_sleep_stats/vmin
			7. Repeat 3~6 for 3times
				Vddmin enter success	Pass

WRD

Test Area		Sub-area	Test Case Name	Procedure	Expect Result	Test Result
Platform	Boot	Boot	EDL download	1. Flash Meta with QFIL(cross test with Fastboot download)	1. Meta flash success, device boot up into UI success	Pass
		Boot	Fastboot download	1. Flash APPS image in fastboot mode(cross test with EDL download, no need test everytime) 2. Run: fastboot reboot	1. Download success	Pass
		Boot	Meta info check	1. cat /vendor/firmware_mnt/verinfo/ver_info.txt	1. Meta info should match	Pass
		Boot	Kernel version check	1. adb shell uname -a	1. Linux Kernel version should be 6.12	Pass
		Boot	Android version check	1. Run: adb root 2. Run: getprop ro.build.version.release	1.Should be "16" (Android W)	Pass

		Boot	Sub-system state	1. Power on device
				2. Read Sub-system list, run:
				  # cat /sys/class/remoteproc/*/name
				3. Read Sub-system status, run:
				  # cat /sys/class/remoteproc/*/state
				4. Check ADSP damon
				  # ps -e | grep rpc
					vienna64:/ # cat /sys/class/remoteproc/remoteproc*/name
					3000000.remoteproc-adsp
					4080000.remoteproc-mss
					32300000.remoteproc-cdsp
					vienna64:/ # cat /sys/class/remoteproc/remoteproc*/state
					running
					running
					running
						Pass

		Boot	M55 uart check	1. Power on device, cmd input: adb wait-for-device root && adb shell "vendor/bin/lpi_utility_test 1 1" 2. Connect UART 3. Select second Port, bitrate 4000000 4. Input "help" and check result 5. Input "build_version swm" to check swm version 6. Input "build_version awm" to check awm version	1. M55 UART is on 2. M55 UART cmd can work 3. M55 swm awm version should match	Pass
		Boot	Themisto boot check	1.Power on device 2.Confirm Themisto is running by: cat /sys/class/coproc/smc_state	1.Themisto is running	Pass
		Boot	Load QCN	1. Start QPST and open "software download" window 2. Choose "Restore" and specify the location of QCN 3. Start the QCN Restore 4. Wait "Memory Restore Completed" untill device will auto restart	1. QCN can flash to "Completed" 2. On normal build, will auto-reboot.	Pass
		Boot	UART console login	1. Open Serial COM tool then connect COM Port on debug board 2. Power on the DUT 3. Check if have log output in serial	1. Serial log output success	Pass
		Boot	adb reboot edl	1. Power on device 2. Run: adb reboot edl	1. Device should enter into 9008 mode	Pass
		Boot	adb reboot bootloader	1. Run: adb reboot bootloader 2. Run: fastboot devices	1. Can enter fastboot mode	Pass
		Boot	fastboot reboot	1. Run: adb reboot bootloader 2. Run: fastboot devices 3. Run: fastboot reboot	1. Can enter fastboot mode and reboot	Pass
		Boot	Factory Reset check	1. adb root 2. adb reboot recovery 3. wipe data/factory reset	1. Factory reset success	Pass
		Boot	adb reboot	1. Power on device 2. Run: adb reboot 3. Repeat 1~2 for 5 times	Reboot success	Pass
		Boot	reboot in shell	1. Power on device 2. Run: adb shell 3. Run: reboot 4. Repeat 1~3 for 5 times	Reboot success	Pass
		Boot	adb shell reboot	1. Power on device 2. Run: adb shell reboot 4. Repeat 1~3 for 5 times	Reboot success	Pass
		Boot	adb remount	1. adb root 2. adb remount 3. adb reboot (to disable dm-verity) 4. adb root 5. adb remount	1. Dm-verity disable success 2. adb remount success	Pass
	Peripherals	Peripherals	Aspen UART	1. Open Serial COM tool then connect COM Port on debug board 2. Power on the DUT	Show log information normal	Pass
		Peripherals	LPI UART	1. Open Serial COM tool then connect COM Port on debug board 2. Power on the DUT	Show log information normal	Pass
		Peripherals	Enter fastboot mode with Volume-	1. Long press Volume- button 2. Connect power cable and USB to power on device 3. Release the Volume- button	Device should enter into fastboot	Pass
		Peripherals	Power key to power on	1. Connect Power only 2. Press the Power button for 3s	Device should boot	Pass
		Peripherals	Power key to enter dump mode	1. Power on the device 2. Long press the Power button for 15s then release the button	1. Device should enter into dump mode 2. QPST should collect dump files	Pass
	Linux Kernel	Dump	Full rampdump collection and parse	1. adb shell, run: # echo c > /proc/sysrq-trigger 2. Open QPST, download dump log, don't use PCATApp 3. Dump log parsing with CrashScope	1. Dump mode enter success 2. Dump log download success && auto-reboot after finish download 3. Dump log parse success (at least no apps mismatch)	Pass
			Dual M55 ramdump collection and parse	1. adb shell, run: # echo c > /proc/sysrq-trigger 2. Open QPST, download dump log, don't use PCATApp 3. Dump log parsing with CrashScope 4. Click CrashScope.Report.csr, check M55 dump under LPAI_SWM and LPAI_AWM	1. After dump parse, can see LPAI_SWM and LPAI_AWM logs	Pass
			Themisto ramdump collection	1. adb shell, run: # echo c > /proc/sysrq-trigger 2. Open QPST, download dump log, don't use PCATApp 3. Check if have Themisto dump logs	1. Themisto dump logs exist with seven SMC binaries: SMC_BT.bin SMC_PERI.bin SMC_SMC.bin SMC_TME.bin SMC_UWB.bin SMC_WLAN_MM.bin SMC_WLAN_PBL.bin	Pass
		Chipset	CPU Info and idle status	1. Run Catch_cpufreq_ftrace.bat script from \\10.64.29.233\Dropbox\2_Wearable\9_SW5100.LAW.2.0\1_Testlimitation\Catch_cpufreq_trace to catch cpufreq ftrace 2. Open below web and open ftrace file on it to get CPU cur frequency: https://ui.perfetto.dev/, use W,A,S,D to control trace view 3. Get CPU online cores # cat /sys/devices/system/cpu/cpu*/online 4. Get CPU max/min frequency # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_max_freq # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_min_freq # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_cur_freq 5. Get CPU governor # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor 6. Or run Chpiset_test.bat script from \\10.64.29.54\Dropbox\1_Robotic\15_QCS8550.LE.1.0\1_TestLimitation\Platform	1. CPU parameters should match design 2. CPU idle freq should be min level	Pass
		Chipset	GPU Info and idle status	1. Run Chpiset_test.bat script \\10.64.29.54\Dropbox\1_Robotic\15_QCS8550.LE.1.0\1_TestLimitation\Platform 2. Get GPU max/min/cur frequency # cat /sys/class/kgsl/kgsl-3d0/devfreq/max_freq # cat /sys/class/kgsl/kgsl-3d0/devfreq/min_freq # cat /sys/class/kgsl/kgsl-3d0/devfreq/cur_freq	1. GPU parameters should match design 2. GPU idle freq should be min level	Pass
		Chipset	DDR Info and idle status	1. Get DDR idle frequency Powe on device. Wait 5 minutes Check available_frequencies? run: # mount -t debugfs none /sys/kernel/debug # cat /sys/devices/system/cpu/bus_dcvs/DDR/available_frequencies | awk 'END{for(i=1;i<=NF;i++) print int($i/1000)}' Check current frequency? run: # cat /sys/kernel/debug/clk/measure_only_mccc_clk/clk_measure 2. Get DDR max frequency # echo 4224000 > /sys/devices/system/cpu/bus_dcvs/DDR/boost_freq # cat /sys/kernel/debug/clk/measure_only_mccc_clk/clk_measure	1. DDR parameters should match design 2. DDR idle freq should be min level	Pass
	Linux Security	Verity	Enable Verity	1. Run: adb root 2. Run: adb enable-verity	1. verity can be enabled success	Pass
		Verity	Disable verity	1. Run: adb root 2. Run: adb disable-verity 3. Run: adb reboot (for settings to take effect) 4. Run: adb root 5. Run: adb disable-verity	1. verity can be disabled success	Pass
		SELinux	SELinux status	1. Get current SELinux status # getenforce	1. Default SELinux status should be Enforcing	Pass

		SELinux	SELinux config	1. Switch from Enforcing to Permissive
				  # setenforce 0
				  # getenforce
				2. Switch from Permissive to Enforcing
				  # setenforce 1
				  # getenforce
					1. SELinux status config to Permissive success
					2. SELinux status config to Enforcing success
						Pass

Security		Security	QSEECOM	1. Run: # qseecom_sample_client v smplap64 1 5	1. QSEECOM run pass	Pass

<table>
<tr><td colspan="2" rowspan="6">Sub-system</td><td>Sub-system state</td><td>Sub-system state</td><td>1. Power on device
2. Read Sub-system list, run:
cat /sys/class/remoteproc/*/name
3. Read Sub-system status, run:
cat /sys/class/remoteproc/*/state
4. Check ADSP damon
ps -e | grep rpc</td><td>1. Sub-system status correct
2. adsprpc cdsproc damon running</td><td>Pass</td></tr>
<tr><td>SSR test on ADSP</td><td>SSR test on ADSP</td><td>adb root
adb remount
adb shell
mount -t debugfs none /sys/kernel/debug
echo enabled > /sys/class/remoteproc/remoteproc0/recovery
Crash inject command:
ADSP QXDM command: send_data 75 37 03 48 02

1. Check SSR count :
cat /sys/class/remoteproc/remoteproc0/device/txn_id
Check SSR log :
dmesg | grep "remoteproc"</td><td>Can see adsp subsystem restart completed info in demsg log</td><td>Pass</td></tr>
<tr><td>SSR test on CDSP</td><td>SSR test on CDSP</td><td>1. Power on device
2. Connect to QXDM
3. Run below command to get the subsystem ID
cat /sys/class/remoteproc/remoteproc2/name
4. Enable SSR
echo enabled > /sys/class/remoteproc/remoteproc2/recovery
5. Send SSR command
send_data 75 37 03 96 00
6. Check SSR count :
cat /sys/class/remoteproc/remoteproc2/device/txn_id
Check SSR log :
dmesg | grep "cdsp is"</td><td>Can see cdsp subsystem restart completed info in demsg log</td><td>Pass</td></tr>
<tr><td>SSR test on MODEM</td><td>SSR test on MODEM</td><td>1. Power on device
2. Connect to QXDM
3. Run below command to get the subsystem ID
cat /sys/class/remoteproc/remoteproc1/name
4. Enable SSR
echo enabled > /sys/class/remoteproc/remoteproc1/recovery
5. In QXDM , send SSR command
send_data 75 37 03 00 00
6. Check SSR count :
cat /sys/class/remoteproc/remoteproc1/device/txn_id
Check SSR log :
dmesg | grep remoteproc and search keywords:"mss is"?dmesg | grep "mpss subsystem is"?</td><td>Can see modem subsystem restart completed info in demsg log</td><td>Pass</td></tr>
<tr><td>SSR test on M55</td><td>SSR test on M55</td><td>adb root
adb remount
adb shell
mount -t debugfs none /sys/kernel/debug
echo 1 > /sys/module/qcom_ramdump/parameters/enable_dump_collection
echo enabled > /sys/class/remoteproc/remoteproc0/recovery
echo enabled > /d/remoteproc/remoteproc0/coredump
echo inline > /d/remoteproc/remoteproc0/coredump
/vendor/bin/subsystem_ramdump &
Crash inject command:
ADSP QXDM command: send_data 75 37 03 48 02
Pull SSR dump from Device instead of PCAT
adb pull /data/vendor/ramdump <destination path></td><td>1.adsp.elf file generated.</td><td>Pass</td></tr>
<tr><td>SSR test on Themisto</td><td>SSR test on Themisto</td><td>1. Trigger Themisto SSR
cat /sys/class/coproc/smc_state
echo enabled > /sys/class/coproc/recovery
echo enabled > /sys/class/coproc/coredump
//command to crash SMC
/vendor/bin/smcpblupdater -c1
cat /sys/class/coproc/smc_state
2. Collect dump
adb root
adb remount
adb pull /data/vendor/ramdump</td><td>1.SMC_-2025-11-22-16-46-24.452.bin file generated.</td><td>Pass</td></tr>
<tr><td colspan="2" rowspan="2">FOTA</td><td rowspan="2">FOTA</td><td>Full-OTA upgrade</td><td>1. Using update package and ota_call.py to get update command:
$ python ota_call.py update.zip for hash output.
2. adb push msm8998-target_files-eng.zip /data/ota_package/update.zip
3. adb root (Don't excute adb remount!!or do adb enable-verity)
4. adb shell
5. Run the upgrade command which get from ota_call.py on step1
eg:
#update_engine_client --update --follow --payload=file:///data/ota_package/update.zip --offset=4682 --size=1110157091 --headers="FILE_HASH=e8U5Q83pMRcu+oiDGjIunuSjKHe76bvAyipPi7Ub3D8=
FILE_SIZE=1110157091
METADATA_HASH=BALg/dKFN/gSfsEntMyo0JM8a7MXm3/a4H3oV0avRA8=
METADATA_SIZE=88800
"

7. In Parallel collect update engine logs.
8. adb logcat | grep -I update_engine
9. After update, reboot request is thrown in logs capture command prompt.
10. Reboot the device.
11. Device will boot from inactive slot with updated build.
12.Check boot slot after OTA
adb shell bootctl get-current-slot</td><td>1. FOTA update should success</td><td>Pass</td></tr>
<tr><td>Ambient mode enter and exit after OTA</td><td>1. Go to Settings -> Display, set AON to ON
2. Set device in Idle: APM off, WLAN connect to AP, USB dis-connected, BT paired with Headset
3. Set time-out as 1min (by default)
4. Back to Watch Face
5. Connect USB/adb
6. Wait to time-out and offload happens
7. Check in logs if status match
8. Click Power key
9. Check if mode exit success
1) UI offload back to Active
2) Have exit info in log
10. Check BT connection during mode exit
11. Check status after mode exit
1) APM off and Network register success
2) WLAN connect to AP</td><td>1. APSS Status should be awake
2. Display offload should work
3. Mode status should be Ambient in logs
4. Mode exit success
5. BT always connected during mode exit
6. Status correct after mode exit</td><td>Pass</td></tr>
<tr><td>MM</td><td rowspan="8">Camera</td><td>Front Camera Preview</td><td>Front _Camera Preview srMax</td><td>1. Open the Camera App and Disable ZSL
2. Take Preview with Selected Resolution, Duration 1min</td><td>1.Camera Preview should be good.
2.On changing the resolution and doing Preview, No ANR or any corruptions should be observed in Preview and Resolution is correct.</td><td>Pass</td></tr>
<tr><td></td><td>Front Camera Snapshot</td><td>Front_Camera ZSL Snapshot srMax12MP</td><td>1. Open the Camera App with zsl on
2. Of Max Resolutions, Take Preview & Snapshot for 3times
3. Pull out the snapshot to PC to check</td><td>1. Camera Preview should be good.
2. On changing the Resolutions and Taking Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview as well as Snapshot. and the resolution is correct
3. The JPEG snapshot and preview should be of the same object of interest. Colors should be reproduced accurately.</td><td>Pass</td></tr>
<tr><td></td><td>Front Camera Live Snapshot</td><td>Front_Camcorder live snapshot_Video_h264_Max_LS_1080p60fps</td><td>1. Open the Camera App.
2. Of required Resolutions, Take live Snapshot during video recording for 3times
3. Pull out to PC to check</td><td>1. Camera Preview should be good.
2. On changing the Resolutions and Taking the live Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview, live snapshot as well as video clip.
3. the resolution of video and the live snapshot is correct
4. The live snapshot and video should be of the same object of interest. Colors should be reproduced accurately.</td><td>Pass</td></tr>
<tr><td></td><td>Front Camera Video_Encode</td><td>Front_Camcorder Recording-Max Resulution/FPS</td><td>1. Launch Camera
2. Switch to camcorder
3. Select settings
4. Select Max resulion 1080p/60fps video quality
5. Start record
6. Click Paust/Start during recording 5 times
7. Stop record</td><td>1. Can video recording
2. File save success and video size and fps is match
3. Can pause/start during playing
4. video playback normally
5. Can hear sound normally</td><td>Pass</td></tr>
<tr><td></td><td>FD</td><td>Front_Camera_Video_Recording_720P30_H264_Duration30s</td><td>1. SD camera app in installed in the devices
2. FD UI is enabled
1. Open Camera app and switch to front camera
2. Switch mode from photo to video
3. Go to settings to enabled FD
4. Towards face and then Set the same video parameters with case described
5. Press start key to video record
6. Wait for duration time, check video recording and generate file</td><td>1. towards face, we can see face ROI in preview
2. video recording will stop automatically at the end of the set time
2. Playback video file by Snapdragon Gallery app, video file duration, fps, quality, resolution are the same with primary set
3. App not find ANR,Crash, Hung</td><td>Pass</td></tr>
<tr><td></td><td>MCTF</td><td>Front_Camera_Video_Recording_720P30_H264_Duration30s</td><td>1. SD camera app in installed in the devices
2.MCTF is enabled
1. Open Camera app and switch to front camera
2. Switch mode from photo to video
3. Enable MCTF
4. Set the same video parameters with case described
5. in the low light condition, press start key to video record
6. Wait for duration time, check video recording and generate file
7.Disable MCTF and repeat the same steps</td><td>1. video recording will stop automatically at the end of the set time
2. Playback video file by Snapdragon Gallery app, video file duration, fps, quality, resolution are the same with primary set
3. App not find ANR,Crash, Hung
4. Less noise in MCTF enabled video clip compared with MCTF disabled video clip</td><td>Pass</td></tr>
<tr><td></td><td>Video HFR - 120FPS</td><td>Front_Camera_Video_Recording_720p120fps_H264_Duration10mins</td><td>1. SD camera app in installed in the devices
2. HFR option enabled in UI
1. Open Camera app
2. Switch mode from photo to HFR
3. HFR record parameter set the same with case name describe
4. Press start key to HFR record
5. Wait for duration time, check video recording and generate file</td><td>1. HFR recording will stop automatically at the end of the set time
2. Playback HFR file by Snapdragon Gallery app, video file duration, encode format, fps, quality, resolution are the same with primary set
3. App not find ANR,Crash, Hung</td><td>Pass</td></tr>
<tr><td></td><td>Shutter sound</td><td>Camera Shutter sound</td><td>1. Take Preview & Snapshot on Front and Back camera, hear camera shutter sound</td><td>1. Can hear camera shutter sound</td><td>Pass</td></tr>
<tr><td></td><td rowspan="4">Video</td><td>Video_Decode</td><td>Video_playback VP9</td><td>1. Push VP9 video file /sdcard/Movies
2. Open Gallery
3. Playback</td><td>Video Playback normally</td><td>Pass</td></tr>
<tr><td></td><td>Video_Decode</td><td>Video_playback HEVC(H265)</td><td>1. Push H265 video file /sdcard/Movies
2. Open Gallery
3. Playback</td><td>Video Playback normally</td><td>Pass</td></tr>
<tr><td></td><td>Video_Decode</td><td>Video_playback H264</td><td>1. Push H264 video file /sdcard/Movies
2. Open Gallery
3. Playback</td><td>Video Playback normally</td><td>Pass</td></tr>
<tr><td></td><td>Video_Decode</td><td>Video_playback Seek/Forward/Back/Volume/Brightness adjust</td><td>1. Playback a video
2. Try Seek, just jump to another timestamp
3. Try Forward/Back, scroll right and left for 3times
4. Try adjust Volume, scroll up and down in right aread for 3times
5. Try adjust Brightness, scroll up and down in left aread for 3times
6. Try adjust Volume via HW Volume up button for 3times</td><td>Seek/Forward/Back/Volume/Brightness adjust works</td><td>Pass</td></tr>
</table>

	Audio	Audio_Encode	AE_encode_by_soundrecord_apk	1. Select AMRNB as a recording format. 2. Start recording by soundrecord apk. 3. Stop the recording. 4. Save the recording.	1. Recorded clip should have some content/data. 2. No pops or glitches should be heard in a recording 3. No crashes observed during the recording. 4. Recording time and recorded clip duration should be the same.	Pass
		Audio_Decode	AD_CompressedOffload_001_wav_playback	1. Play wav > 60s using Music app	1. Verify that compressed-offload clip plays with audio heard all the time and no crashes are seen in the logs 2. Verify in logcat for "out_set_parameters" being set to "compress-offload-playback"	Pass
		Audio_Decode	AD_CompressedOffload_001_mp3_playback	1. Play mp3 > 60s using Music app	1. Verify that compressed-offload clip plays with audio heard all the time and no crashes are seen in the logs 2. Verify in logcat for "out_set_parameters" being set to "compress-offload-playback"	Pass
		Audio_Decode	AD_CompressedOffload_001_aac_playback	1. Play aac > 60 s using Music app	1. Verify that compressed-offload clip plays with audio heard all the time and no crashes are seen in the logs 2. Verify in logcat for "out_set_parameters" being set to "compress-offload-playback"	Pass
		Audio_Decode	AD_musicapp_playback_hw_volume_profile	1. Playback audio using Music app 2. Adjust volume from mute to max and back to mute by HW volume buttons 3. Verify if playback volume changed during audio playback	1. No crashes or ANRs 2. No loss in audio or glich/pops. 3. Volume displays an exponential profile, with same levels as PCM. 4. Volume level display moves accordingly. 5. Soft stepping is enabled 6. No saturation at max volumes	Pass
		Speaker_Protection	AD_FBSP_quick_calibration_noplayback	1.adb pull /vendor/etc/audio/sku_vienna64/resourcemanager_vienna.xml <param key="logging_level" value ="7" /> (change 3 to 7) <speaker_protection_enabled>1</speaker_protection_enabled> (change 0 to 1) <quick_cal_time>90</quick_cal_time> (change 90) adb push C:\Users\shafeng\resourcemanager_vienna.xml /vendor/etc/audio/sku_vienna64/resourcemanager_vienna.xml adb reboot wait 90s # adb shell logcat -v threadtime | grep -sn "SpeakerProtection" ? to check log kyewords "SpeakerProtection"? #cd /mnt/vendor/persist/hlos_rfs/shared #ls -l and check keywords" audio.cal"	1. Verify calibration success in logcat log(adb logcat | grep SpeakerProtection) 2. Verify calibration file(audio.cal) generated in file system	Pass
		Speaker_Protection	AD_FBSP_audio_playback_speaker	1. Playback music after setting below steps	1. No audible pops/clicks heard	Pass
		Fluence	VoIP_MO_Call_with_Fluence_enabled_DMEF	1.Run C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\Audio\Fluence\push_voip_dmef_xml.bat 2.install weixin.apk 3.Open QACT Tool,and connect device 4. make a MO Wechat voice call 5.Check if the QACT tool displays the word 'fluence'	The sound output correct with no noise	Pass
		Fluence	VoIP_MT_Call_with_Fluence_enabled_DMEF	1.Run C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\Audio\Fluence\push_voip_dmef_xml.bat adb reboot 2.install weixin.apk 3.Open QACT Tool,and connect device 4. make a MT Wechat voice call 5.Check if the QACT tool displays the word 'fluence'	The sound output correct with no noise	Pass
		Haptics	AD_Haptics_with_Alarm	1. Enable haptics in settings 2. Set Offload app alarm with some time 3. Obeserve alarm with haptics	1. Verify haptics with Offload app alarm	Pass
	Display	Display	DI_Brightness	1. Open the Down bar 2. Change the Brightness 3. Check the brightness is changed	Brightness can be changed	Pass
		DI_Scrolling	Check if scrolling is even. Settings -> Scroll -> wait 2 Sec. for fallback	Scrolling down should be free of Janks or stutters.	Scrolling is even	Pass
	Touch	Touch	Single tap	1.Touch to open a app 2.Touch to scroll the app menu	Touch works smoothly	Pass
Sensor		Physical Sensor	Accel	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Place DUT for UP/DOWN/LEFT/RIGHT/FRONT/BACK six different orientation	1. Accel(x, y, z) data should corresponding to the place orientation 2. For example, when place UP, should be (0, 0, 9.8)	Pass
		Physical Sensor	Gyro	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Rotation DUT on YAW/ROLL/PITCH three different orientation	1. Gyro(x, y,z) data change should corresponding to the rotation orientation 2. For example, when rotation on YAW, only Z data change	Pass
		Physical Sensor	Mag	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Place DUT on different orientation	Mag(x, y,z) data should corresponding to the place orientation	Pass
		Physical Sensor	ALS/Light	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Use hand to cover and then left the DUT	Light data should decrease and the increase	Pass
		Physical Sensor	Pressure	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type	Pressure data should be reasonable	Pass
		Physical Sensor	Skin temperature	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Use hand to click Skin temperature sensor	Skin temperature data should have change when click by hand	Pass
		Physical Sensor	ENG	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Use hand to click ENG sensor	ENG data should have change when click by hand	Pass
Connectivity	Co-work	WLAN/BT Co-work	WLAN+BT enable at same time	1. Open wifi menu 2. Enable bt and pair with headset for audio playback 3. Open wifi menu 4. Enable wifi and connect AP to ping 5. Check the BT/Wlan all open at same time	WLAN+BT co-work success	Pass
	WLAN	STA	STA: Connect to an Ref-AP with Password	1. Connect to an AP which has password setup (WPA2) 2. Enter the password and ensure that connection can be established.	1. Wifi can enable success 2. Wifi ssid will show in list 3. Input password can connect success	Pass
		STA	STA: Browse the web or Ping a Web site	1. Open Browser 2. Input a Web link 3. Or ping a Web site like: www.baidu.com	1. Connect "Hydra" success 2. Can open the browser and view the web page	Pass
		Remember	STA: Remember Network	1. After connecting to a network, select "Remember network" 2. Disable WLAN and then Enable WLAN for 3times 3. Ensure that the network which was saved earlier still shows up.	Auto connect success	Pass
	BT	Pair	BT Device Pair	1. Switch ON BT. 2. Search for BT Devices. 3. Pair to the Devices. 4. Pair and unpair BT 3 times	1. BT enable success 2. Show search BT devices list 3. Can pair the any devices	Pass
		A2DP	Playback over BT Headset	1. Switch on BT 2. Search BT Headset Devices. 3. Bond to an Identified BT Headset Device. (A2DP) 4. Start MP3 Playback and check for Audio Playback over the headset. Check for subjective audio quality. 5. Increase/decrease volume via DUT HW button and BT Volume button during BT Headset Device 6. Repeat steps 5 for 3times via DUT HW button and BT Headset button	1. BT enable success 2. Show search BT devices list 3. Can pair the any devices 4. Audio can playback normally via headset 5. Volume can be adjusted	Pass
		AVRCP	Pause/Start/Next/Last over BT Headset	1. Pair on BT headset and play audio 2. On BT headset, click Pause/Start/Next/Last sound via APP in DUT side for 3times 3. On BT headset, click Pause/Start/Next/Last sound via Headset button for 3times	1. Pause/Start/Next/Last works well via app and BT Headset button control	Pass
		Device Switch	DeviceSwitch_Playback_speaker_BT	1. Playback over BT Headset 2. Swith BT Headset Devices to Speaker via APP in DUT side, 3. Restart audio playback when swith to device. 4. And then swith Speaker to BT Headset via BT Headset button 5. Repeat 1~4 for 5times	1. BT/Speaker switch success via APP and BT Headset button control 2. No delay in voice recognition after swithed	Pass
		Rememer	BT Device Pair remember	1. Disable BT 2. Then enable BT 3. Enable/Disable BT for 3 times 4. Wait 10s to check if BT can auto pair to device	Auto pair success	Pass
	UWB	UWB	UWB Enable	1. Power on device 2. You can enable UWB by UI: Settings--Connected devices--Connection preferences--UWB 3. Or you can trigger enablement of UWB via shell command force-country-code? adb shell "cmd uwb force-country-code enabled US" cat uwb state: adb shell cmd uwb status enable uwb: adb shell cmd uwb enable-uwb	1. UWB is enable	Pass
	NFC	Enable/Disable	NFC enable and Disable	1. Go to Menu -->Settings--> NFC 2. Turn on NFC and disable/enable for 3times	1. NFC enable/disable success	Pass
		NFC tag read	NFC tag read	1. Install NFC tag apk: \\jianwu22\Dropbox\2_Wearable\11_SW6100.LAW.1.0\01.TestLimitation\NFC\NFC Tools_v8.6_apkpure.com 1.apk 2. Turn on NFC and enable it 3. Open NFC tag app, read tag	1. NFC tag can be read	Pass
Telephony		IMS register	Register IMS	1. Inset a SIM card support VoLTE or VoINR	1. When IMS register success, should have HD tag	Pass
		VoLTE	VoLTE_MO answer	1.Camp to VoLTE network and register IMS 2.Make MO Call 3.After MO call get connected Make MT call 4.Device reject the MT call 5.Device answer the MT call	1.Mo call get connected, the first active call should hold.	Pass
		VoLTE	VoLTE_MT anwser	1.Camp to VoLTE network and register IMS 2.Make MT Call 3.After MT call get connected Make MO call	1.devices can reject the MT call, and the first active call should keep . 2.device can answer the MT call , the first call hold.	Pass
		VoNR	VoNR_MO answer	1. Camp to NR network and register IMS 2. Make MO Call	1. When IMS register success, should have HD tag 2. MO call should connect, during the call, network mode should still be 5G 3. Data still works using Ping	Pass
		VoNR	VoNR_MT answer	1. Camp to NR network and register IMS 2. Make MT Call	1. When IMS register success, should have HD tag 2. MO call should connect, during the call, network mode should still be 5G 3. Data still works using Ping	Pass
		SMS VoNR	SMS VoNR	1. Send MO SMS to other devices	1. Mo SMS send from DUT must be receveid .	Pass
		SMS VoNR	SMS VoNR	1. Make MT SMS from other device to DUT	1. DUT should receive MT SMS .	Pass
		Data-Celluar	LTE(4G)_Browse	1.Do browsing	1. Data browsing should happen	Pass
		Data-Celluar	NR(5G)_Browse	1.Do browsing	1. Data browsing should happen	Pass
		Adversial-APM	Adversial-APM_ON-OFF	1.Put dut in APM, dial any non emergency number 2.Off the APM, Make MO call	1. dial shouldn’t work, it should show to dial take dut out of APM 2.Call should get connected.	Pass
Location		GPS	Standalone app tracking	1. Install GPS-test Apk on the target 2. Open the GPS Test App. 3. Test outside, wait 1~2min	1. GPS/beidou/GLONSS satellites all can be found 2.Latitude/Longitude position info can be gived	Pass

Wear Tech	UI	Date/time auto sync by WWAN	1. Register to NR network 2. Check if date/time can auto sync	1. Date/time can auto sync	Pass
		Watch Face and Sports UI switch in tile	1.Enable the UI Swiping by: adb root; adb remount adb shell setprop persist.watchui.quickSettingsEnabled true reboot 2.Left swipe to sports UI from watch face UI Right swipe to watch face UI from sports UI 3. repeat 5 times	1.Switch the UI successfully, the UI content is correct	Pass
		Watch Face switch	1. Go to Face setting, choose digital watch face 2. Switch to analog watch face 3. Switch to digital watch face with sports 4. Switch to analog watch face with sports	Each watch face switch success, and UI display normal	Pass
		Enter Quick settings panel	1.Slide from top to bottom on the watchface 2.Check the Quick settings panel display, include: 1) 5 buttons: WIFI ON/OFF, AirplaneON/OFF, Bluetooth ON/OFF, Settings, Brightness 2) 3 icons: Battery status,Telecommunications signal, Time	Enter Successfully and Display correctly 1.Buttons: Include a pattern in the middle with a circle on the outside Icons: 1) Battery status(display charging icon while charging or display remaining battery level when not charge) 2) Telecom signal icon(display signal status bright or dark, show signal type like LTE,UMTS and so on upper left of the icon 3) Time and date (displays the current hours, minutes, seconds, day of the week, and date)	Pass
		Click the Settings button	1.Click the Settings button to jump to the Setting page	1.Jump to the Settings page successfully	Pass
		Restart	1. Long press power key or go to Settings -> Restart 2. Perform restart and observe the device behavior	Restart function and UI works well	Pass
		Power off	1. Long press power key or go to Settings -> Power Off 2. Perform power off and observe the device behavior	Power off function and UI works well	Pass
		Factory Data Reset	1. Go to Menu -->Settings--> Factory Reset 2. Perform factory data reset and observe the device behavior	1. All user related data should be removed and device should be as it was after first boot up.	Pass
	Offload	Ambient mode enter and exit	1. Go to Settings -> Display, set AON to ON 2. Set device in Idle: APM off, WLAN connect to AP, BT paired with Headset 3. Set time-out as 15sec (by default) 4. Back to Watch Face 5. Connect USB/adb 6. Wait to time-out and offload happens 7. Check if display offload work 8. Check in logs if status match 9. Click Power key 10. Check if mode exit success 1) UI offload back to Active 2) Have exit info in log 11. Check BT connection during mode exit 12. Check status after mode exit 1) APM off and Network register success 2) WLAN connect to AP	1. Display offload should work 3. Mode status should match in logs 4. Mode exit success 5. BT always connected during mode exit 6. Status correct after mode exit	Pass
		AON on under Ambient mode	1. Go to Settings -> Display, set AON to ON 2. Set device in Idle: APM off, WLAN connect to AP, BT paired with Headset 3. Set time-out as 1min (by default) 4. Back to Watch Face 5. Wait to time-out and offload happens	1. Display offload should work and display should be in always-on mode	Pass
		AON off under Ambient mode	1. Go to Settings -> Display, set AON to OFF 2. Set device in Idle: APM off, WLAN connect to AP, BT paired with Headset 3. Set time-out as 1min (by default) 4. Back to Watch Face 5. Wait to time-out and offload happens	1. Display offload should work and display should be in off state	Pass
		Reboot under ambient mode	1. Enter Ambient mode 2. Reboot DUT under Ambient mode 3. Repeat 5 times	1. Reboot success under Ambient mode	Pass
ML	QNN	QNN on NSP	C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\QNNTest\QNN\qnn-sdk-v2.37.1.25072\Push_QNN.bat adb push lib\aarch64-android\. /vendor/lib/ adb push bin\aarch64-android\. /vendor/bin/ adb push lib\hexagon-v81\unsigned\. /vendor/lib/rfsa/adsp/ adb push \\lab8510\Dropbox\AISW_132328\qnn-test-v2.37.0.25_Test_pack\qnn-test-v2.37.0.250724175447_124859\model_tf2_10_custom_mlperf_deeplabv3_qat\inputs\. /data/inputs adb push \\lab8510\Dropbox\AISW_132328\qnn-test-v2.37.0.25_Test_pack\qnn-test-v2.37.0.250724175447_124859\model_tf2_10_custom_mlperf_deeplabv3_qat\quantized\aarch64-android\tf2_10_custom_mlperf_deeplabv3_qat_quantized.so /vendor/ Executed command: # cd /data/inputs # qnn-net-run --log_level=error --profiling_level=basic --backend=/vendor/lib/libQnnHtp.so --input_list=/data/inputs/input_list.txt --output_dir=/data/inputs/ --model=/vendor/tf2_10_custom_mlperf_deeplabv3_qat_quantized.so Get Inference stats: # qnn-profile-viewer --input_log /data/inputs/qnn-profiling-data.log	Execute Stats (Average): ------------------------ Total Inference Time: --------------------- Graph 0 (qnn_model): NetRun: 14210 us Backend (Number of HVX threads used): 4 count Backend (RPC (execute) time): 13181 us Backend (QNN accelerator (execute) time): 11184 us Backend (Accelerator (execute) time): 10994 us Backend (Accelerator (execute excluding wait) time): 10324 us Backend (QNN (execute) time): 14381 us	Pass
Product	Power	Suspend check with SIM card and APM OFF	1. Insert SIM card and Power on device # adb shell "mount -t debugfs none /sys/kernel/debug" 2. adb root and run below commands # cat /d/suspend_stats 3. Disconnect USB and wait for time-out or use power key to suspend 4. After display cut off, wait 30s 5. Connect USB or use power key to resume 6. Check if enter suspend success # cat /d/suspend_stats 7. Repeat 3~6 for 3times	Suspend enter success	Pass
		Suspend check with SIM card and APM ON	1. Insert SIM card and Power on device, then open Airplane mode # adb shell "mount -t debugfs none /sys/kernel/debug" 2. adb root and run below commands # cat /d/suspend_stats 3. Disconnect USB and wait for time-out or use power key to suspend 4. After display cut off, wait 30s 5. Connect USB or use power key to resume 6. Check if enter suspend # cat /d/suspend_stats 7. Repeat 3~6 for 3times	Suspend enter success	Pass
		Vddmin check with SIM card and APM OFF	1. Insert SIM card and Power on device # adb shell "mount -t debugfs none /sys/kernel/debug" 2. adb root and run below commands # cat /d/suspend_stats # cat /d/qcom_stats/aosd # cat /d/qcom_stats/cxsd 3. Disconnect USB and wait for time-out or use power key to suspend 4. After display cut off, wait 30s 5. Connect USB or use power key to resume 6. Check if enter suspend/vddmin success # cat /d/suspend_stats # cat /d/qcom_stats/aosd # cat /d/qcom_stats/cxsd 7. Repeat 3~6 for 3times	Vddmin enter success	Pass
		Vddmin check with SIM card and APM ON	1. Insert SIM card and Power on device, then open Airplane mode # adb shell "mount -t debugfs none /sys/kernel/debug" 2. adb root and run below commands # cat /d/suspend_stats # cat /sys/kernel/debug/qcom_sleep_stats/vmin 3. Disconnect USB and wait for time-out or use power key to suspend 4. After display cut off, wait 30s 5. Connect USB or use power key to resume 6. Check if enter suspend/vddmin success # cat /d/suspend_stats # cat /sys/kernel/debug/qcom_sleep_stats/vmin 7. Repeat 3~6 for 3times	Vddmin enter success	Pass

APQ

Test Area	Sub-area	Test Case Name	Procedure	Expect Result	Test Result

Platform	Boot	Boot	EDL download	1. Flash Meta with QFIL(cross test with Fastboot download)	1. Meta flash success, device boot up into UI success	Pass
		Boot	Fastboot download	1. Flash APPS image in fastboot mode(cross test with EDL download, no need test everytime) 2. Run: fastboot reboot	1. Download success	Pass
		Boot	Meta info check	1. cat /vendor/firmware_mnt/verinfo/ver_info.txt	1. Meta info should match	Pass
		Boot	Kernel version check	1. adb shell uname -a	1. Linux Kernel version should be 6.12	Pass
		Boot	Android version check	1. Run: adb root 2. Run: getprop ro.build.version.release	1.Should be "16" (Android W)	Pass
		Boot	Sub-system state	1. Power on device 2. Read Sub-system list, run: # cat /sys/class/remoteproc/*/name 3. Read Sub-system status, run: # cat /sys/class/remoteproc/*/state 4. Check ADSP damon # ps -e | grep rpc	vienna64:/ # cat /sys/class/remoteproc/remoteproc*/name 3000000.remoteproc-adsp 4080000.remoteproc-mss 32300000.remoteproc-cdsp vienna64:/ # cat /sys/class/remoteproc/remoteproc*/state running running running	Pass
		Boot	M55 uart check	1. Power on device, cmd input: adb wait-for-device root && adb shell "vendor/bin/lpi_utility_test 1 1" 2. Connect UART 3. Select second Port, bitrate 4000000 4. Input "help" and check result 5. Input "build_version swm" to check swm version 6. Input "build_version awm" to check awm version	1. M55 UART is on 2. M55 UART cmd can work 3. M55 swm awm version should match	Pass
		Boot	Themisto boot check	1.Power on device 2.Confirm Themisto is running by: cat /sys/class/coproc/smc_state	1.Themisto is running	Pass
		Boot	Load QCN	1. Start QPST and open "software download" window 2. Choose "Restore" and specify the location of QCN 3. Start the QCN Restore 4. Wait "Memory Restore Completed" untill device will auto restart	1. QCN can flash to "Completed" 2. On normal build, will auto-reboot.	Pass
		Boot	UART console login	1. Open Serial COM tool then connect COM Port on debug board 2. Power on the DUT 3. Check if have log output in serial	1. Serial log output success	Pass
		Boot	adb reboot edl	1. Power on device 2. Run: adb reboot edl	1. Device should enter into 9008 mode	Pass
		Boot	adb reboot bootloader	1. Run: adb reboot bootloader 2. Run: fastboot devices	1. Can enter fastboot mode	Pass
		Boot	fastboot reboot	1. Run: adb reboot bootloader 2. Run: fastboot devices 3. Run: fastboot reboot	1. Can enter fastboot mode and reboot	Pass
		Boot	Factory Reset check	1. adb root 2. adb reboot recovery 3. wipe data/factory reset	1. Factory reset success	Pass
		Boot	adb reboot	1. Power on device 2. Run: adb reboot 3. Repeat 1~2 for 5 times	Reboot success	Pass
		Boot	reboot in shell	1. Power on device 2. Run: adb shell 3. Run: reboot 4. Repeat 1~3 for 5 times	Reboot success	Pass
		Boot	adb shell reboot	1. Power on device 2. Run: adb shell reboot 4. Repeat 1~3 for 5 times	Reboot success	Pass
		Boot	adb remount	1. adb root 2. adb remount 3. adb reboot (to disable dm-verity) 4. adb root 5. adb remount	1. Dm-verity disable success 2. adb remount success	Pass
	Peripherals	Peripherals	Aspen UART	1. Open Serial COM tool then connect COM Port on debug board 2. Power on the DUT	Show log information normal	Pass
		Peripherals	LPI UART	1. Open Serial COM tool then connect COM Port on debug board 2. Power on the DUT	Show log information normal	Pass
		Peripherals	Enter fastboot mode with Volume-	1. Long press Volume- button 2. Connect power cable and USB to power on device 3. Release the Volume- button	Device should enter into fastboot	Pass
		Peripherals	Power key to power on	1. Connect Power only 2. Press the Power button for 3s	Device should boot	Pass
		Peripherals	Power key to enter dump mode	1. Power on the device 2. Long press the Power button for 15s then release the button	1. Device should enter into dump mode 2. QPST should collect dump files	Pass
	Linux Kernel	Dump	Full rampdump collection and parse	1. adb shell, run: # echo c > /proc/sysrq-trigger 2. Open QPST, download dump log, don't use PCATApp 3. Dump log parsing with CrashScope	1. Dump mode enter success 2. Dump log download success && auto-reboot after finish download 3. Dump log parse success (at least no apps mismatch)	Pass
			Dual M55 ramdump collection and parse	1. adb shell, run: # echo c > /proc/sysrq-trigger 2. Open QPST, download dump log, don't use PCATApp 3. Dump log parsing with CrashScope 4. Click CrashScope.Report.csr, check M55 dump under LPAI_SWM and LPAI_AWM	1. After dump parse, can see LPAI_SWM and LPAI_AWM logs	Pass
			Themisto ramdump collection	1. adb shell, run: # echo c > /proc/sysrq-trigger 2. Open QPST, download dump log, don't use PCATApp 3. Check if have Themisto dump logs	1. Themisto dump logs exist with seven SMC binaries: SMC_BT.bin SMC_PERI.bin SMC_SMC.bin SMC_TME.bin SMC_UWB.bin SMC_WLAN_MM.bin SMC_WLAN_PBL.bin	Pass
		Chipset	CPU Info and idle status	1. Run Catch_cpufreq_ftrace.bat script from \\10.64.29.233\Dropbox\2_Wearable\9_SW5100.LAW.2.0\1_Testlimitation\Catch_cpufreq_trace to catch cpufreq ftrace 2. Open below web and open ftrace file on it to get CPU cur frequency: https://ui.perfetto.dev/, use W,A,S,D to control trace view 3. Get CPU online cores # cat /sys/devices/system/cpu/cpu*/online 4. Get CPU max/min frequency # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_max_freq # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_min_freq # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_cur_freq 5. Get CPU governor # cat /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor 6. Or run Chpiset_test.bat script from \\10.64.29.54\Dropbox\1_Robotic\15_QCS8550.LE.1.0\1_TestLimitation\Platform	1. CPU parameters should match design 2. CPU idle freq should be min level	Pass
		Chipset	GPU Info and idle status	1. Run Chpiset_test.bat script \\10.64.29.54\Dropbox\1_Robotic\15_QCS8550.LE.1.0\1_TestLimitation\Platform 2. Get GPU max/min/cur frequency # cat /sys/class/kgsl/kgsl-3d0/devfreq/max_freq # cat /sys/class/kgsl/kgsl-3d0/devfreq/min_freq # cat /sys/class/kgsl/kgsl-3d0/devfreq/cur_freq	1. GPU parameters should match design 2. GPU idle freq should be min level	Pass
		Chipset	DDR Info and idle status	1. Get DDR idle frequency Powe on device. Wait 5 minutes Check available_frequencies? run: # mount -t debugfs none /sys/kernel/debug # cat /sys/devices/system/cpu/bus_dcvs/DDR/available_frequencies | awk 'END{for(i=1;i<=NF;i++) print int($i/1000)}' Check current frequency? run: # cat /sys/kernel/debug/clk/measure_only_mccc_clk/clk_measure 2. Get DDR max frequency # echo 4224000 > /sys/devices/system/cpu/bus_dcvs/DDR/boost_freq # cat /sys/kernel/debug/clk/measure_only_mccc_clk/clk_measure	1. DDR parameters should match design 2. DDR idle freq should be min level	Pass
	Linux Security	Verity	Enable Verity	1. Run: adb root 2. Run: adb enable-verity	1. verity can be enabled success	Pass
		Verity	Disable verity	1. Run: adb root 2. Run: adb disable-verity 3. Run: adb reboot (for settings to take effect) 4. Run: adb root 5. Run: adb disable-verity	1. verity can be disabled success	Pass
		SELinux	SELinux status	1. Get current SELinux status # getenforce	1. Default SELinux status should be Enforcing	Pass
		SELinux	SELinux config	1. Switch from Enforcing to Permissive # setenforce 0 # getenforce 2. Switch from Permissive to Enforcing # setenforce 1 # getenforce	1. SELinux status config to Permissive success 2. SELinux status config to Enforcing success	Pass
Security		Security	QSEECOM	1. Run: # qseecom_sample_client v smplap64 1 5	1. QSEECOM run pass	Pass
Sub-system		Sub-system state	Sub-system state	1. Power on device 2. Read Sub-system list, run: # cat /sys/class/remoteproc/*/name 3. Read Sub-system status, run: # cat /sys/class/remoteproc/*/state 4. Check ADSP damon # ps -e | grep rpc	1. Sub-system status correct 2. adsprpc cdsproc damon running	Pass
		SSR test on ADSP	SSR test on ADSP	adb root adb remount adb shell mount -t debugfs none /sys/kernel/debug echo enabled > /sys/class/remoteproc/remoteproc0/recovery Crash inject command: ADSP QXDM command: send_data 75 37 03 48 02 1. Check SSR count : cat /sys/class/remoteproc/remoteproc0/device/txn_id Check SSR log : dmesg | grep "remoteproc"	Can see adsp subsystem restart completed info in demsg log	Pass
		SSR test on CDSP	SSR test on CDSP	1. Power on device 2. Connect to QXDM 3. Run below command to get the subsystem ID # cat /sys/class/remoteproc/remoteproc2/name 4. Enable SSR # echo enabled > /sys/class/remoteproc/remoteproc2/recovery 5. Send SSR command send_data 75 37 03 96 00 6. Check SSR count : cat /sys/class/remoteproc/remoteproc2/device/txn_id Check SSR log : dmesg | grep "cdsp is"	Can see cdsp subsystem restart completed info in demsg log	Pass
		SSR test on MODEM	SSR test on MODEM	1. Power on device 2. Connect to QXDM 3. Run below command to get the subsystem ID # cat /sys/class/remoteproc/remoteproc1/name 4. Enable SSR # echo enabled > /sys/class/remoteproc/remoteproc1/recovery 5. In QXDM , send SSR command send_data 75 37 03 00 00 6. Check SSR count : cat /sys/class/remoteproc/remoteproc1/device/txn_id Check SSR log : dmesg | grep "mss is"?dmesg | grep "mpss subsystem is"?	Can see modem subsystem restart completed info in demsg log	Pass
		SSR test on M55	SSR test on M55	adb root adb remount adb shell mount -t debugfs none /sys/kernel/debug echo 1 > /sys/module/qcom_ramdump/parameters/enable_dump_collection echo enabled > /sys/class/remoteproc/remoteproc0/recovery echo enabled > /d/remoteproc/remoteproc0/coredump echo inline > /d/remoteproc/remoteproc0/coredump /vendor/bin/subsystem_ramdump & Crash inject command: ADSP QXDM command: send_data 75 37 03 48 02 Pull SSR dump from Device instead of PCAT adb pull /data/vendor/ramdump <destination path>	1.adsp.elf file generated.	Pass
		SSR test on Themisto	SSR test on Themisto	1. Trigger Themisto SSR cat /sys/class/coproc/smc_state echo enabled > /sys/class/coproc/recovery echo enabled > /sys/class/coproc/coredump //command to crash SMC /vendor/bin/smcpblupdater -c1 cat /sys/class/coproc/smc_state 2. Collect dump adb root adb remount adb pull /data/vendor/ramdump	1.SMC_-2025-11-22-16-46-24.452.bin file generated.	Pass

FOTA	FOTA	Full-OTA upgrade	1. Using update package and ota_call.py to get update command:
			    $ python ota_call.py update.zip for hash output.
			2. adb push msm8998-target_files-eng.zip /data/ota_package/update.zip
			3. adb root (Don't excute adb remount!!or do adb enable-verity)
			4. adb shell
			5. Run the upgrade command which get from ota_call.py on step1
			eg:
			#update_engine_client --update --follow --payload=file:///data/ota_package/update.zip --offset=4682 --size=1110157091 --headers="FILE_HASH=e8U5Q83pMRcu+oiDGjIunuSjKHe76bvAyipPi7Ub3D8=
			FILE_SIZE=1110157091
			METADATA_HASH=BALg/dKFN/gSfsEntMyo0JM8a7MXm3/a4H3oV0avRA8=
			METADATA_SIZE=88800
			"
			
			7. In Parallel collect update engine logs.
			8. adb logcat | grep -I update_engine
			9. After update, reboot request is thrown in logs capture command prompt.
			10. Reboot the device.
			11. Device will boot from inactive slot with updated build.
			12.Check boot slot after OTA
			    # adb shell bootctl get-current-slot
				1. FOTA update should success	Pass

		Ambient mode enter and exit after OTA	1. Go to Settings -> Display, set AON to ON
			2. Set device in Idle: APM off, WLAN connect to AP, USB dis-connected, BT paired with Headset
			3. Set time-out as 1min (by default)
			4. Back to Watch Face
			5. Connect USB/adb
			6. Wait to time-out and offload happens
			7. Check in logs if status match
			8. Click Power key
			9. Check if mode exit success
			    1) UI offload back to Active
			    2) Have exit info in log
			10. Check BT connection during mode exit
			11. Check status after mode exit
			    1) APM off and Network register success
			    2) WLAN connect to AP
				1. APSS Status should be awake
				2. Display offload should work
				3. Mode status should be Ambient in logs
				4. Mode exit success
				5. BT always connected during mode exit
				6. Status correct after mode exit
					Pass

MM	Camera	Front Camera Preview	Front _Camera Preview srMax	1. Open the Camera App and Disable ZSL 2. Take Preview with Selected Resolution, Duration 1min	1.Camera Preview should be good. 2.On changing the resolution and doing Preview, No ANR or any corruptions should be observed in Preview and Resolution is correct.	Pass
		Front Camera Snapshot	Front_Camera ZSL Snapshot srMax12MP	1. Open the Camera App with zsl on 2. Of Max Resolutions, Take Preview & Snapshot for 3times 3. Pull out the snapshot to PC to check	1. Camera Preview should be good. 2. On changing the Resolutions and Taking Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview as well as Snapshot. and the resolution is correct 3. The JPEG snapshot and preview should be of the same object of interest. Colors should be reproduced accurately.	Pass
		Front Camera Live Snapshot	Front_Camcorder live snapshot_Video_h264_Max_LS_1080p60fps	1. Open the Camera App. 2. Of required Resolutions, Take live Snapshot during video recording for 3times 3. Pull out to PC to check	1. Camera Preview should be good. 2. On changing the Resolutions and Taking the live Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview, live snapshot as well as video clip. 3. the resolution of video and the live snapshot is correct 4. The live snapshot and video should be of the same object of interest. Colors should be reproduced accurately.	Pass
		Front Camera Video_Encode	Front_Camcorder Recording-Max Resulution/FPS	1. Launch Camera 2. Switch to camcorder 3. Select settings 4. Select Max resulion 1080p/60fps video quality 5. Start record 6. Click Paust/Start during recording 5 times 7. Stop record	1. Can video recording 2. File save success and video size and fps is match 3. Can pause/start during playing 4. video playback normally 5. Can hear sound normally	Pass
		Back Camera Preview	Back_Camera Preview srMax	1. Open the Camera App and Disable ZSL 2. Take Preview with Selected Resolution, Duration 1min	1. Camera Preview should be good. 2. On changing the resolution and doing Preview, No ANR or any corruptions should be observed in Preview and Resolution is correct.	Pass
		Back Camera Snapshot	Back_Camera ZSL Snapshot srMax12MP	1. Open the Camera App with zsl on 2. Of Max Resolutions, Take Preview & Snapshot for 3times 3. pull out the snapshot to PC to check	1. Camera Preview should be good. 2. On changing the Resolutions and Taking Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview as well as Snapshot. and the resolution is correct 3.The JPEG snapshot and preview should be of the same object of interest. Colors should be reproduced accurately.	Pass
		Back Camera Live Snapshot	Back_Camcorder live snapshot_Video_h265_Max_LS_1080p60fps	1. Open the Camera App. 2. Of required Resolutions, Take live Snapshot during video recording for 3times 3. Pull out to PC to check	1. Camera Preview should be good. 2. On changing the Resolutions and Taking the live Snapshots; the application should be handled gracefully and No ANR or any corruptions should be observed in Preview, live snapshot as well as video clip. 3. the resolution of video and the live snapshot is correct The live snapshot and video should be of the same object of interest. Colors should be reproduced accurately.	Pass
		Back Camera Video_Encode	Back_Camcorder Recording-Max Resulution/FPS	1. Launch Camera 2. Switch to camcorder 3. Select settings 4. Select Max resulion1080p/60fps video quality 5. Start record 6. Click Paust/Start during recording 5 times 7. Stop record	1. Can video recording 2. File save success and video size and fps is match 3. Can pause/start during playing 4. video playback normally	Pass
		FD	Front_Camera_Video_Recording_720P30_H264_Duration30s	1. SD camera app in installed in the devices 2. FD UI is enabled 1. Open Camera app and switch to front camera 2. Switch mode from photo to video 3. Go to settings to enabled FD 4. Towards face and then Set the same video parameters with case described 5. Press start key to video record 6. Wait for duration time, check video recording and generate file	1. towards face, we can see face ROI in preview 2. video recording will stop automatically at the end of the set time 2. Playback video file by Snapdragon Gallery app, video file duration, fps, quality, resolution are the same with primary set 3. App not find ANR,Crash, Hung	Pass
		MCTF	Front_Camera_Video_Recording_720P30_H264_Duration30s	1. SD camera app in installed in the devices 2.MCTF is enabled 1. Open Camera app and switch to front camera 2. Switch mode from photo to video 3. Enable MCTF 4. Set the same video parameters with case described 5. in the low light condition, press start key to video record 6. Wait for duration time, check video recording and generate file 7.Disable MCTF and repeat the same steps	1. video recording will stop automatically at the end of the set time 2. Playback video file by Snapdragon Gallery app, video file duration, fps, quality, resolution are the same with primary set 3. App not find ANR,Crash, Hung 4. Less noise in MCTF enabled video clip compared with MCTF disabled video clip	Pass
		Video HFR - 120FPS	Front_Camera_Video_Recording_720p120fps_H264_Duration10mins	1. SD camera app in installed in the devices 2. HFR option enabled in UI 1. Open Camera app 2. Switch mode from photo to HFR 3. HFR record parameter set the same with case name describe 4. Press start key to HFR record 5. Wait for duration time, check video recording and generate file	1. HFR recording will stop automatically at the end of the set time 2. Playback HFR file by Snapdragon Gallery app, video file duration, encode format, fps, quality, resolution are the same with primary set 3. App not find ANR,Crash, Hung	Pass
		Video EIS	Front_Camera_VideoRecording_720P30_H264_DurationNolimit_EIS on	1. SD camera app in installed in the devices 2. image Stabilization setting is functional in SD camera to enable EIS 3. Video EIS setting is functional in SD camera to enable EIS v3 4. Dual TFE based 1080p + 50% margin for video output from camera is enabled 1. Open Camera app? 2. Switch mode from photo to video 3. Record parameter set the same with case name describe, enable EIS by set image Stabilization on, keep device on repeat Slight shaking, press start key to video record, wait for duration time. 4. Disable EIS by set image Stabilization off, , keep device on repeat Slight shaking, press start key to video record, wait for duration time. 5.Pull out and check video recording and generate file by PC		Pass
		Shutter sound	Camera Shutter sound	1. Take Preview & Snapshot on Front and Back camera, hear camera shutter sound	1. Can hear camera shutter sound	Pass
	Video	Video_Decode	Video_playback VP9	1. Push VP9 video file /sdcard/Movies 2. Open Gallery 3. Playback	Video Playback normally	Pass
		Video_Decode	Video_playback HEVC(H265)	1. Push H265 video file /sdcard/Movies 2. Open Gallery 3. Playback	Video Playback normally	Pass
		Video_Decode	Video_playback H264	1. Push H264 video file /sdcard/Movies 2. Open Gallery 3. Playback	Video Playback normally	Pass
		Video_Decode	Video_playback Seek/Forward/Back/Volume/Brightness adjust	1. Playback a video 2. Try Seek, just jump to another timestamp 3. Try Forward/Back, scroll right and left for 3times 4. Try adjust Volume, scroll up and down in right aread for 3times 5. Try adjust Brightness, scroll up and down in left aread for 3times 6. Try adjust Volume via HW Volume up button for 3times	Seek/Forward/Back/Volume/Brightness adjust works	Pass
	Audio	Audio_Encode	AE_encode_by_soundrecord_apk	1. Select AMRNB as a recording format. 2. Start recording by soundrecord apk. 3. Stop the recording. 4. Save the recording.	1. Recorded clip should have some content/data. 2. No pops or glitches should be heard in a recording 3. No crashes observed during the recording. 4. Recording time and recorded clip duration should be the same.	Pass
		Audio_Decode	AD_CompressedOffload_001_wav_playback	1. Play wav > 60s using Music app	1. Verify that compressed-offload clip plays with audio heard all the time and no crashes are seen in the logs 2. Verify in logcat for "out_set_parameters" being set to "compress-offload-playback"	Pass
		Audio_Decode	AD_CompressedOffload_001_mp3_playback	1. Play mp3 > 60s using Music app	1. Verify that compressed-offload clip plays with audio heard all the time and no crashes are seen in the logs 2. Verify in logcat for "out_set_parameters" being set to "compress-offload-playback"	Pass
		Audio_Decode	AD_CompressedOffload_001_aac_playback	1. Play aac > 60 s using Music app	1. Verify that compressed-offload clip plays with audio heard all the time and no crashes are seen in the logs 2. Verify in logcat for "out_set_parameters" being set to "compress-offload-playback"	Pass
		Audio_Decode	AD_musicapp_playback_hw_volume_profile	1. Playback audio using Music app 2. Adjust volume from mute to max and back to mute by HW volume buttons 3. Verify if playback volume changed during audio playback	1. No crashes or ANRs 2. No loss in audio or glich/pops. 3. Volume displays an exponential profile, with same levels as PCM. 4. Volume level display moves accordingly. 5. Soft stepping is enabled 6. No saturation at max volumes	Pass
		Speaker_Protection	AD_FBSP_quick_calibration_noplayback	1.adb pull /vendor/etc/audio/sku_vienna64/resourcemanager_vienna.xml <param key="logging_level" value ="7" /> (change 3 to 7) <speaker_protection_enabled>1</speaker_protection_enabled> (change 0 to 1) <quick_cal_time>90</quick_cal_time> (change 90) adb push C:\Users\shafeng\resourcemanager_vienna.xml /vendor/etc/audio/sku_vienna64/resourcemanager_vienna.xml adb reboot wait 90s # adb shell logcat -v threadtime | grep -sn "SpeakerProtection" ? to check log kyewords "SpeakerProtection"? #cd /mnt/vendor/persist/hlos_rfs/shared #ls -l and check keywords" audio.cal"	1. Verify calibration success in logcat log(adb logcat | grep SpeakerProtection) 2. Verify calibration file(audio.cal) generated in file system	Pass
		Speaker_Protection	AD_FBSP_audio_playback_speaker	1. Playback music after setting below steps	1. No audible pops/clicks heard	Pass
		Fluence	VoIP_MO_Call_with_Fluence_enabled_DMEF	1.Run C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\Audio\Fluence\push_voip_dmef_xml.bat 2.install weixin.apk 3.Open QACT Tool,and connect device 4. make a MO Wechat voice call 5.Check if the QACT tool displays the word 'fluence'	The sound output correct with no noise	Pass
		Fluence	VoIP_MT_Call_with_Fluence_enabled_DMEF	1.Run C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\Audio\Fluence\push_voip_dmef_xml.bat adb reboot 2.install weixin.apk 3.Open QACT Tool,and connect device 4. make a MT Wechat voice call 5.Check if the QACT tool displays the word 'fluence'	The sound output correct with no noise	Pass
		Haptics	AD_Haptics_with_Alarm	1. Enable haptics in settings 2. Set Offload app alarm with some time 3. Obeserve alarm with haptics	1. Verify haptics with Offload app alarm	Pass
	Display	Display	DI_Brightness	1. Open the Down bar 2. Change the Brightness 3. Check the brightness is changed	Brightness can be changed	Pass
		DI_Scrolling	Check if scrolling is even. Settings -> Scroll -> wait 2 Sec. for fallback	Scrolling down should be free of Janks or stutters.	Scrolling is even	Pass
	Touch	Touch	Single tap	1.Touch to open a app 2.Touch to scroll the app menu	Touch works smoothly	Pass

Sensor		Physical Sensor	Accel	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Place DUT for UP/DOWN/LEFT/RIGHT/FRONT/BACK six different orientation	1. Accel(x, y, z) data should corresponding to the place orientation 2. For example, when place UP, should be (0, 0, 9.8)	Pass
		Physical Sensor	Gyro	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Rotation DUT on YAW/ROLL/PITCH three different orientation	1. Gyro(x, y,z) data change should corresponding to the rotation orientation 2. For example, when rotation on YAW, only Z data change	Pass
		Physical Sensor	Mag	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Place DUT on different orientation	Mag(x, y,z) data should corresponding to the place orientation	Pass
		Physical Sensor	ALS/Light	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Use hand to cover and then left the DUT	Light data should decrease and the increase	Pass
		Physical Sensor	Pressure	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type	Pressure data should be reasonable	Pass
		Physical Sensor	Skin temperature	Sensor test first choose QSensorTest app, 1. Open QsensorTest app and click correspond sensor type 2. Use hand to click Skin temperature sensor	Skin temperature data should have change when click by hand	Pass
Connectivity	Co-work	WLAN/BT Co-work	WLAN+BT enable at same time	1. Open wifi menu 2. Enable bt and pair with headset for audio playback 3. Open wifi menu 4. Enable wifi and connect AP to ping 5. Check the BT/Wlan all open at same time	WLAN+BT co-work success	Pass
	WLAN	STA	STA: Connect to an Ref-AP with Password	1. Connect to an AP which has password setup (WPA2) 2. Enter the password and ensure that connection can be established.	1. Wifi can enable success 2. Wifi ssid will show in list 3. Input password can connect success	Pass
		STA	STA: Browse the web or Ping a Web site	1. Open Browser 2. Input a Web link 3. Or ping a Web site like: www.baidu.com	1. Connect "Hydra" success 2. Can open the browser and view the web page	Pass
		Remember	STA: Remember Network	1. After connecting to a network, select "Remember network" 2. Disable WLAN and then Enable WLAN for 3times 3. Ensure that the network which was saved earlier still shows up.	Auto connect success	Pass
	BT	Pair	BT Device Pair	1. Switch ON BT. 2. Search for BT Devices. 3. Pair to the Devices. 4. Pair and unpair BT 3 times	1. BT enable success 2. Show search BT devices list 3. Can pair the any devices	Pass
		A2DP	Playback over BT Headset	1. Switch on BT 2. Search BT Headset Devices. 3. Bond to an Identified BT Headset Device. (A2DP) 4. Start MP3 Playback and check for Audio Playback over the headset. Check for subjective audio quality. 5. Increase/decrease volume via DUT HW button and BT Volume button during BT Headset Device 6. Repeat steps 5 for 3times via DUT HW button and BT Headset button	1. BT enable success 2. Show search BT devices list 3. Can pair the any devices 4. Audio can playback normally via headset 5. Volume can be adjusted	Pass
		AVRCP	Pause/Start/Next/Last over BT Headset	1. Pair on BT headset and play audio 2. On BT headset, click Pause/Start/Next/Last sound via APP in DUT side for 3times 3. On BT headset, click Pause/Start/Next/Last sound via Headset button for 3times	1. Pause/Start/Next/Last works well via app and BT Headset button control	Pass
		Device Switch	DeviceSwitch_Playback_speaker_BT	1. Playback over BT Headset 2. Swith BT Headset Devices to Speaker via APP in DUT side, 3. Restart audio playback when swith to device. 4. And then swith Speaker to BT Headset via BT Headset button 5. Repeat 1~4 for 5times	1. BT/Speaker switch success via APP and BT Headset button control 2. No delay in voice recognition after swithed	Pass
		Rememer	BT Device Pair remember	1. Disable BT 2. Then enable BT 3. Enable/Disable BT for 3 times 4. Wait 10s to check if BT can auto pair to device	Auto pair success	Pass
	UWB	UWB	UWB Enable	1. Power on device 2. You can enable UWB by UI: Settings--Connected devices--Connection preferences--UWB 3. Or you can trigger enablement of UWB via shell command force-country-code? adb shell "cmd uwb force-country-code enabled US" cat uwb state: adb shell cmd uwb status enable uwb: adb shell cmd uwb enable-uwb	1. UWB is enable	Pass
	NFC	Enable/Disable	NFC enable and Disable	1. Go to Menu -->Settings--> NFC 2. Turn on NFC and disable/enable for 3times	1. NFC enable/disable success	Pass
		NFC tag read	NFC tag read	1. Install NFC tag apk: \\jianwu22\Dropbox\2_Wearable\11_SW6100.LAW.1.0\01.TestLimitation\NFC\NFC Tools_v8.6_apkpure.com 1.apk 2. Turn on NFC and enable it 3. Open NFC tag app, read tag	1. NFC tag can be read	Pass
Location		GPS	Standalone app tracking	1. Install GPS-test Apk on the target 2. Open the GPS Test App. 3. Test outside, wait 1~2min	1. GPS/beidou/GLONSS satellites all can be found 2.Latitude/Longitude position info can be gived	Pass
Wear Tech		UI	Watch Face and Sports UI switch in tile	1.Enable the UI Swiping by: adb root; adb remount adb shell setprop persist.watchui.quickSettingsEnabled true reboot 2.Left swipe to sports UI from watch face UI Right swipe to watch face UI from sports UI 3. repeat 5 times	1.Switch the UI successfully, the UI content is correct	Pass
			Watch Face switch	1. Go to Face setting, choose digital watch face 2. Switch to analog watch face 3. Switch to digital watch face with sports 4. Switch to analog watch face with sports	Each watch face switch success, and UI display normal	Pass
			Enter Quick settings panel	1.Slide from top to bottom on the watchface 2.Check the Quick settings panel display, include: 1) 5 buttons: WIFI ON/OFF, AirplaneON/OFF, Bluetooth ON/OFF, Settings, Brightness 2) 3 icons: Battery status,Telecommunications signal, Time	Enter Successfully and Display correctly 1.Buttons: Include a pattern in the middle with a circle on the outside Icons: 1) Battery status(display charging icon while charging or display remaining battery level when not charge) 2) Telecom signal icon(display signal status bright or dark, show signal type like LTE,UMTS and so on upper left of the icon 3) Time and date (displays the current hours, minutes, seconds, day of the week, and date)	Pass
			Click the Settings button	1.Click the Settings button to jump to the Setting page	1.Jump to the Settings page successfully	Pass
			Restart	1. Long press power key or go to Settings -> Restart 2. Perform restart and observe the device behavior	Restart function and UI works well	Pass
			Power off	1. Long press power key or go to Settings -> Power Off 2. Perform power off and observe the device behavior	Power off function and UI works well	Pass
			Factory Data Reset	1. Go to Menu -->Settings--> Factory Reset 2. Perform factory data reset and observe the device behavior	1. All user related data should be removed and device should be as it was after first boot up.	Pass
		Offload	Ambient mode enter and exit	1. Go to Settings -> Display, set AON to ON 2. Set device in Idle: APM off, WLAN connect to AP, BT paired with Headset 3. Set time-out as 15sec (by default) 4. Back to Watch Face 5. Connect USB/adb 6. Wait to time-out and offload happens 7. Check if display offload work 8. Check in logs if status match 9. Click Power key 10. Check if mode exit success 1) UI offload back to Active 2) Have exit info in log 11. Check BT connection during mode exit 12. Check status after mode exit 1) APM off and Network register success 2) WLAN connect to AP	1. Display offload should work 3. Mode status should match in logs 4. Mode exit success 5. BT always connected during mode exit 6. Status correct after mode exit	Pass
			AON on under Ambient mode	1. Go to Settings -> Display, set AON to ON 2. Set device in Idle: APM off, WLAN connect to AP, BT paired with Headset 3. Set time-out as 1min (by default) 4. Back to Watch Face 5. Wait to time-out and offload happens	1. Display offload should work and display should be in always-on mode	Pass
			AON off under Ambient mode	1. Go to Settings -> Display, set AON to OFF 2. Set device in Idle: APM off, WLAN connect to AP, BT paired with Headset 3. Set time-out as 1min (by default) 4. Back to Watch Face 5. Wait to time-out and offload happens	1. Display offload should work and display should be in off state	Pass
			Reboot under ambient mode	1. Enter Ambient mode 2. Reboot DUT under Ambient mode 3. Repeat 5 times	1. Reboot success under Ambient mode	Pass
ML		QNN	QNN on NSP	C:\IOT\Wearable\SW6100.LAW.1.0\TestLimitation\QNNTest\QNN\qnn-sdk-v2.37.1.25072\Push_QNN.bat adb push lib\aarch64-android\. /vendor/lib/ adb push bin\aarch64-android\. /vendor/bin/ adb push lib\hexagon-v81\unsigned\. /vendor/lib/rfsa/adsp/ adb push \\lab8510\Dropbox\AISW_132328\qnn-test-v2.37.0.25_Test_pack\qnn-test-v2.37.0.250724175447_124859\model_tf2_10_custom_mlperf_deeplabv3_qat\inputs\. /data/inputs adb push \\lab8510\Dropbox\AISW_132328\qnn-test-v2.37.0.25_Test_pack\qnn-test-v2.37.0.250724175447_124859\model_tf2_10_custom_mlperf_deeplabv3_qat\quantized\aarch64-android\tf2_10_custom_mlperf_deeplabv3_qat_quantized.so /vendor/ Executed command: # cd /data/inputs # qnn-net-run --log_level=error --profiling_level=basic --backend=/vendor/lib/libQnnHtp.so --input_list=/data/inputs/input_list.txt --output_dir=/data/inputs/ --model=/vendor/tf2_10_custom_mlperf_deeplabv3_qat_quantized.so Get Inference stats: # qnn-profile-viewer --input_log /data/inputs/qnn-profiling-data.log	Execute Stats (Average): ------------------------ Total Inference Time: --------------------- Graph 0 (qnn_model): NetRun: 14210 us Backend (Number of HVX threads used): 4 count Backend (RPC (execute) time): 13181 us Backend (QNN accelerator (execute) time): 11184 us Backend (Accelerator (execute) time): 10994 us Backend (Accelerator (execute excluding wait) time): 10324 us Backend (QNN (execute) time): 14381 us	Pass
Product		Power	Suspend check with SIM card and APM OFF	1. Insert SIM card and Power on device # adb shell "mount -t debugfs none /sys/kernel/debug" 2. adb root and run below commands # cat /d/suspend_stats 3. Disconnect USB and wait for time-out or use power key to suspend 4. After display cut off, wait 30s 5. Connect USB or use power key to resume 6. Check if enter suspend success # cat /d/suspend_stats 7. Repeat 3~6 for 3times	Suspend enter success	Pass
			Suspend check with SIM card and APM ON	1. Insert SIM card and Power on device, then open Airplane mode # adb shell "mount -t debugfs none /sys/kernel/debug" 2. adb root and run below commands # cat /d/suspend_stats 3. Disconnect USB and wait for time-out or use power key to suspend 4. After display cut off, wait 30s 5. Connect USB or use power key to resume 6. Check if enter suspend # cat /d/suspend_stats 7. Repeat 3~6 for 3times	Suspend enter success	Pass
			Vddmin check with SIM card and APM OFF	1. Insert SIM card and Power on device # adb shell "mount -t debugfs none /sys/kernel/debug" 2. adb root and run below commands # cat /d/suspend_stats # cat /d/qcom_stats/aosd # cat /d/qcom_stats/cxsd 3. Disconnect USB and wait for time-out or use power key to suspend 4. After display cut off, wait 30s 5. Connect USB or use power key to resume 6. Check if enter suspend/vddmin success # cat /d/suspend_stats # cat /d/qcom_stats/aosd # cat /d/qcom_stats/cxsd 7. Repeat 3~6 for 3times	Vddmin enter success	Pass
			Vddmin check with SIM card and APM ON	1. Insert SIM card and Power on device, then open Airplane mode # adb shell "mount -t debugfs none /sys/kernel/debug" 2. adb root and run below commands # cat /d/suspend_stats # cat /sys/kernel/debug/qcom_sleep_stats/vmin 3. Disconnect USB and wait for time-out or use power key to suspend 4. After display cut off, wait 30s 5. Connect USB or use power key to resume 6. Check if enter suspend/vddmin success # cat /d/suspend_stats # cat /sys/kernel/debug/qcom_sleep_stats/vmin 7. Repeat 3~6 for 3times	Vddmin enter success	Pass

9. References

Title	Number
Qualcomm Technologies, Inc.	
Qualcomm Product Configuration Assistant Tool (PCAT) User Guide	80-PR518-1
Qualcomm Unified Tools Service (QUTS) User Guide	80-PG522-3
QXDM Professional User Guide	80-V1241-21
QXDM PROFESSIONAL LINUX EDITION USER GUIDE	80-V1241-22
Qualcomm Hexagon Tools Installation Guide	80-N2040-32
SW6100 Software User Guide	80-74841-50

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